

Research Portfolio

Junyoung Park

Research Achievements



Junyoung Park

➤ Education

- M.S. (2022.03 ~ 2024.02) : Chung-Ang Univ. Civil Engineering (Advisor: Prof. Jongwoong Park) GPA: 4.39/4.5
- B.S. (2017.03 ~ 2022.02) : Chung-Ang Univ. Civil & Environmental Plant Engineering, Mathematics (Double Major) GPA: 4.2/4.5

➤ Experience

- UDNS Co., Ltd. (2024.05 ~ Present): Researcher, R&D Division
 - Bridge Co., Ltd. (2023.01 ~ 2024.02): Inside Director
- Civil & Environmental Plant Eng. Major GPA: 4.23/4.5
Mathematics Major GPA: 4.18/4.5

➤ English Proficiency

– TOEFL 100 (C1): Reading 5.0, Listening 4.5, Writing 5.0, Speaking 5.0 (issued 2026.02.28)

➤ Key Awards: 5

Competition	Award	Date	Topic	Awarded by
2023 Fall Conference	Excellent Paper Presentation Award	2024.04.04	Scalable IoT Sensor and Cloud Computing Framework for Continuous Bridge Monitoring	Korea Institute for Structural Maintenance and Inspection
2022 Korea Future Talent Festival	Gold Prize	2022.11.25	IoT-Based Smart Monitoring System for Stability and Quality Control of Prefab Members in Transit	National Research Foundation of Korea
12th LH Land Development Technology Awards	Excellence Award	2021.11.25	Citizen-Participatory Smart City via QR-Code Visualization of Sensor Data	Korea Land & Housing Corporation
Smart Construction Challenge 2021 (IoT·AI·Sensing)	Korea National Railway Chairman's Award	2021.11.17	IoT Sensor and Cloud-Based Continuous Bridge Displacement Monitoring System	Ministry of Land, Infrastructure and Transport
2019 Smart Earthquake Response Idea Contest	Minister of the Interior and Safety Award	2019.07.25	Earthquake Alert System Using Multisensory Media and Smart Sensors	Ministry of the Interior and Safety

Research Achievements

➤ Papers: 12 (SCI 7, Non-SCI 3, Under Review 2)

Type	Published	Title	Journal / Publisher	Authors
SCI (Under review)	2026	Development of a Portable Multimetric Sensing System for Bridge Load Testing and Load-Carrying Capacity Evaluation	Measurement	J. Park, J. Shin, J. Won
	2026	Low power multi-channel strain sensing and instant shape visualization for digital twin	Journal of Civil Structural Health Monitoring	S. Khan, J. Park, J. Park, I. Kim, J. Won
SCI	2024	Development of AI-Based Multisensory System for Monitoring Quay Wall Events	Journal of Marine Science and Engineering	J. Shin, J. Park, J. Won, J. Park, J. Min
	2024	Tendon Stress Estimation from Strain Data of a Bridge Girder Using Machine Learning-Based Surrogate Model	Sensors	SU Khayam, A Ajmal, J Park, IH Kim, JW Park
	2023	Monitoring Precast Structures during Transportation Using a Portable Sensing System	Automation in Construction	S.U. Khayam, J. Won, J. Shin, J. Park, J.W. Park
	2021	Development of low-cost wireless sensing system for smart ultra-high performance concrete	Sensors	H.V. Le, T. Kim, S. Khan, J. Park, S. Kim, Y. Jang, D. Kim
	2021	Development of a reference-free indirect bridge displacement sensing system	Sensors	J. Won, J.W. Park, J. Park, J. Shin, M. Park
	2021	SSVM: An ultra-low-power strain sensing and visualization module for long-term structural health monitoring	Sensors	S. Khan, J. Won, J. Shin, J. Park, S. Kim, D. Kim
	2021	BLESeis: Low-cost IOT sensor for smart earthquake detection and notification	Sensors	J. Won, J. Park, J.W. Park
Non-SCI	2025	YOLO-Based Automatic Detection of Load Segments in High-Speed Weigh-in-Motion Systems	KSCE J. Civ. Environ. Eng. Res.	J. Park, J.W. Kim, Y. Cho, Y. Jung
	2024	Ensuring the Stability of an In-Motion Dynamic State Measurement System for Freight Trucks	J. Korean Soc. Mech. Technol.	J. Park, J.S. Kim, J.W. Kim, Y. Cho
	2021	Low-Power IoT Sensor and Cloud-Based Data Fusion Displacement Measurement for Continuous Bridge Monitoring	J. Comput. Struct. Eng. Inst. Korea	J. Park, J. Shin, J. Won, J.W. Park, M. Park

✓ Google Scholar: <https://scholar.google.com/citations?hl=en&user=Y00UTgQAAAAJ>

Research Achievements

➤ Conferences: 9 (International: 3, Domestic: 6)

Type	Presented	Title	Organization	Authors
International	2024.05.20	QR SENSOR: A citizen-assisted, QR-based sensory data acquisition and cloud computing approach for Structural Health Monitoring	EMI/EMC 2024	J.W. Park, J. Park , C. Min
	2023.08.23	Cloud-based Ambient Bridge Monitoring using Strain and Acceleration Measurements	Korean Institute of BIM	J. Park , J. Shin, J.W. Park
	2022.11.18	Development of IoT Sensor and Cloud-based Server for Cloud-based Bridge Long-term Monitoring	KICEM	J. Park , J. Shin, J.W. Park
Domestic	2025.11.14	Development of a Cloud-Based Integrated Automated Enforcement Platform	KSCE	J. Park , J.W. Kim, J.H. Kim
	2023.10.12	Scalable IoT Sensor and Cloud Computing Framework for Continuous Bridge Monitoring	KSMI	J. Park , J. Shin, J.W. Park
	2023.09.18	Long-Term Bridge Monitoring Based on IoT Sensors and Cloud Computing	Korean Soc. Bridge & Struct. Eng.	J. Park
	2022.06.30	Cloud-Based IoT Sensor System for Bridge Load-Carrying Capacity Evaluation	KSPHM	J.W. Park, J. Park , J. Shin, J. Won
	2022.05.26	Optimization of Acceleration-Strain-Based Displacement Estimation Considering Bridge Boundary Conditions	KSNVE	J. Park , J. Shin, J.W. Park
	2021.04.08	Cloud-Based Bridge Displacement Monitoring System	COSEIK	J. Park , J. Shin, J. Won, J.W. Park, M. Park

➤ Domestic Patents: 9 (Registered 7, Filed 1)

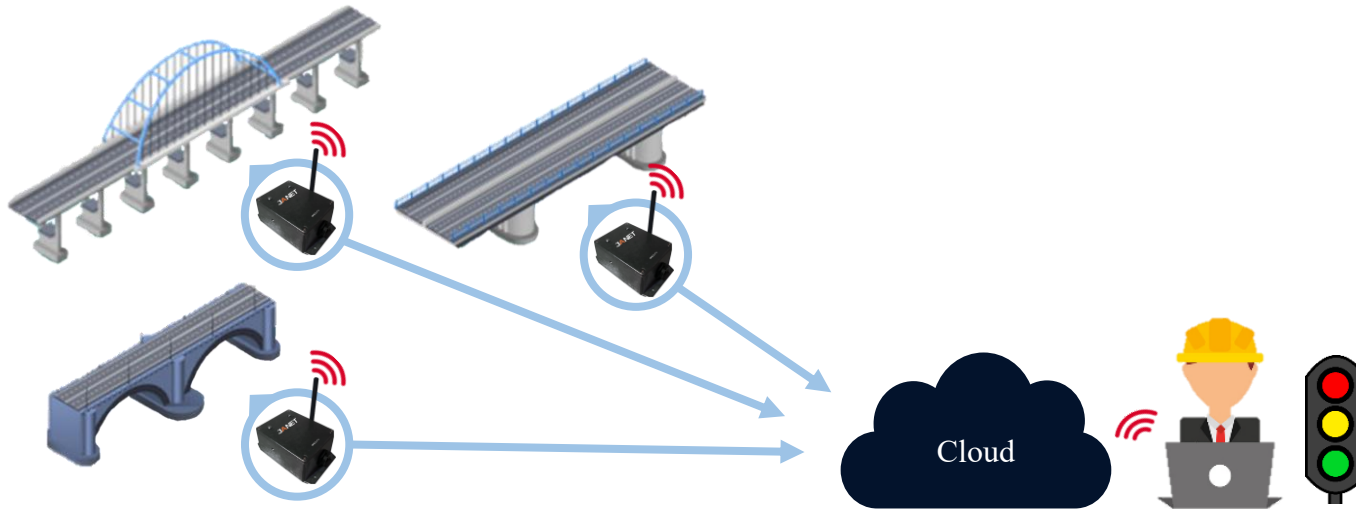
Type	Reg. (Appl.) No.	Title of Invention	Inventors
To Be Filed	–	Heterogeneous Sensor-Based Vehicle Dimension Measurement System for Screening Oversized Vehicles	J.W. Kim, Y. Cho, Y. Jung, J. Park
Filed	10-2025-0174354	System and Method for Continuous Bridge Load-Carrying Capacity Evaluation Using High-Speed WIM	J.W. Kim, Y. Cho, J. Park
Filed	10-2025-0068214	FIR Filter-Based Method and Apparatus for Real-Time Correction of Dynamic Freight Truck Weight	J.W. Kim, Y. Cho, J. Park
Registered	10-2853678	System and Method for Automatic Recognition and Analysis of Load Segments Using IoT Sensors for Structural Load-Carrying Capacity Evaluation	J.W. Park, J. Park , J. Shin
Registered	10-2892584	Concrete Crack Trajectory Monitoring Device Using a Resistive Touchpad and Long-Term Measurement History Visualization System	J.W. Park, K. Kim, J. Park
Registered	10-2558124	IoT Sensor-Based Integrated Monitoring System and Method for Precast Member Transport	J.W. Park, J. Park , J. Shin, J. Lee
Registered	10-2705761	Smart IoT Sensor Data Preprocessing Method for Structural Monitoring Using Cloud Computing, and Deep Learning-Based Automated Anomaly Detection System	J.W. Park, J. Won, J. Park , J. Shin
Registered	10-2389222	Citizen-Participatory Smart City System Using Sensor Measurement and Data Visualization	S. Park, J. Shin, J. Park
Registered	10-2353297	Smart IoT Seismic Sensor Unit and Alert Method Using the Same	J.W. Park, J. Park

Research Projects

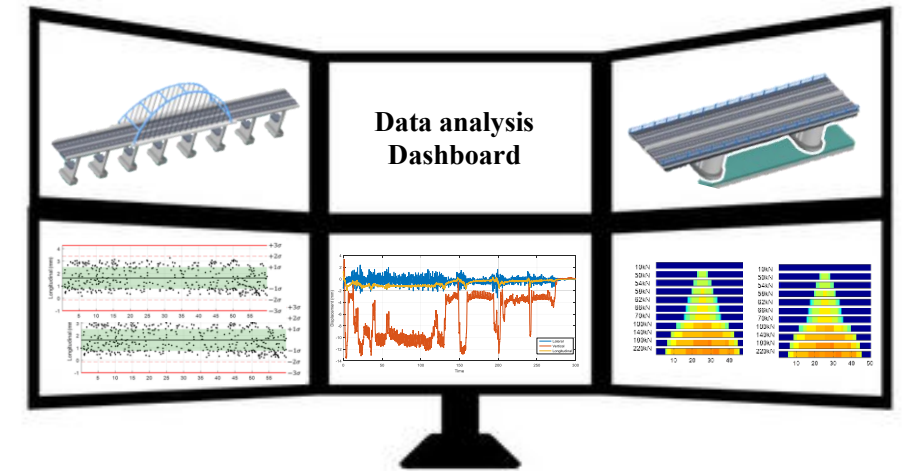
Research Projects (1)

Project ①: IoT Sensing and Cloud System for Smart Bridge Level 3 (MOLIT, 2021.04 ~ 2022.12)

- Bridge safety is ensured through in-depth safety inspections, but labor-intensive load-carrying capacity evaluation cannot keep pace with a **shrinking construction workforce** and a **growing number of aging bridges**. In addition, the long inspection interval of 6 months to 2 years makes it difficult to respond quickly to problems in bridges.
- To address this, a **smart sensor** that is Stand-Alone, requiring no separate power or communication unit, was developed to **measure acceleration and strain**, the key indicators of bridge safety; **a cloud server was built** to collect sensor data automatically and unattended. In addition, **a DB and built-in algorithms** were deployed in the cloud so that **FFT, displacement estimation**, and other computations are performed automatically on the measured data.
- As a result, a system was developed that uses **IoT sensors** and **a cloud system** to **monitor bridge safety** continuously over the long term.



1. Smart Sensor



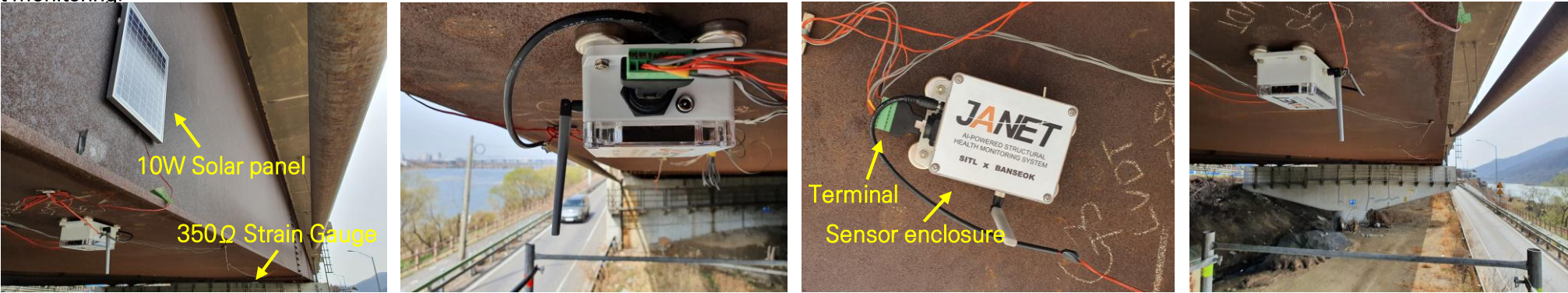
2. Cloud System

Research Projects (1)

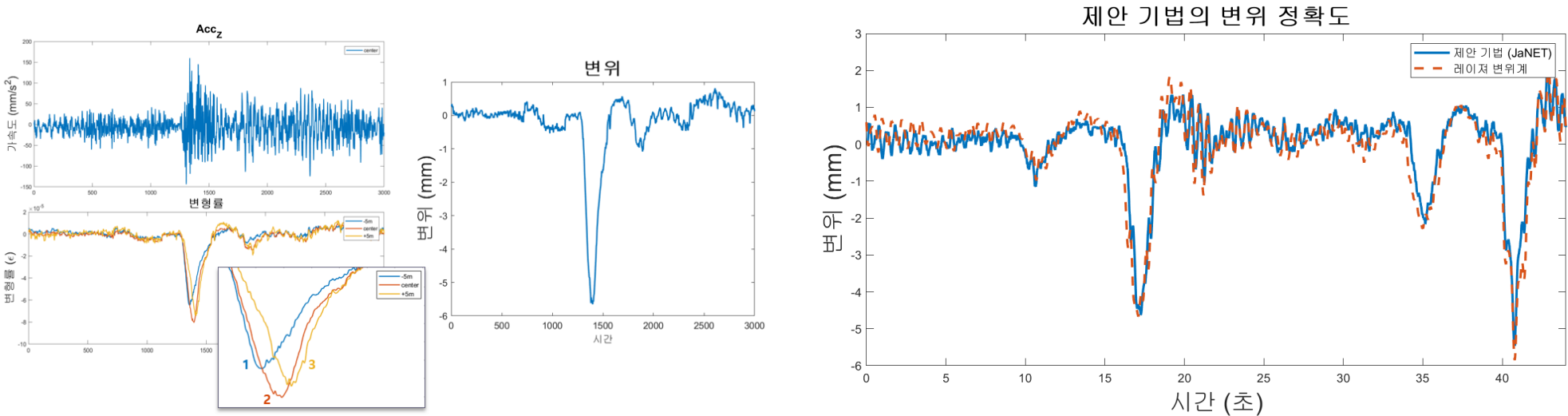
Project ①: IoT Sensing and Cloud System for Smart Bridge Level 3 (MOLIT, 2021.04 ~ 2022.12)

- Going beyond the lab, the system was applied to real bridge sites, with sensors installed on the Paldang Overpass (Gyeonggi-do) and Cheongdam Bridge No. 1 (Seoul) for long-term displacement monitoring.

[Paldang Overpass Installation]



[Measured Data]



Research Projects (1)

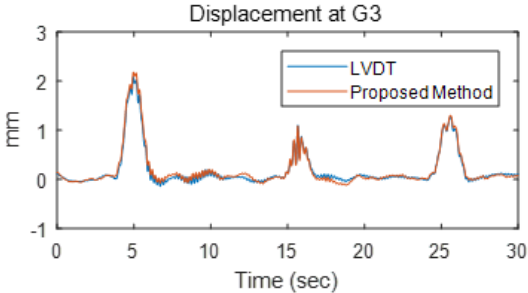
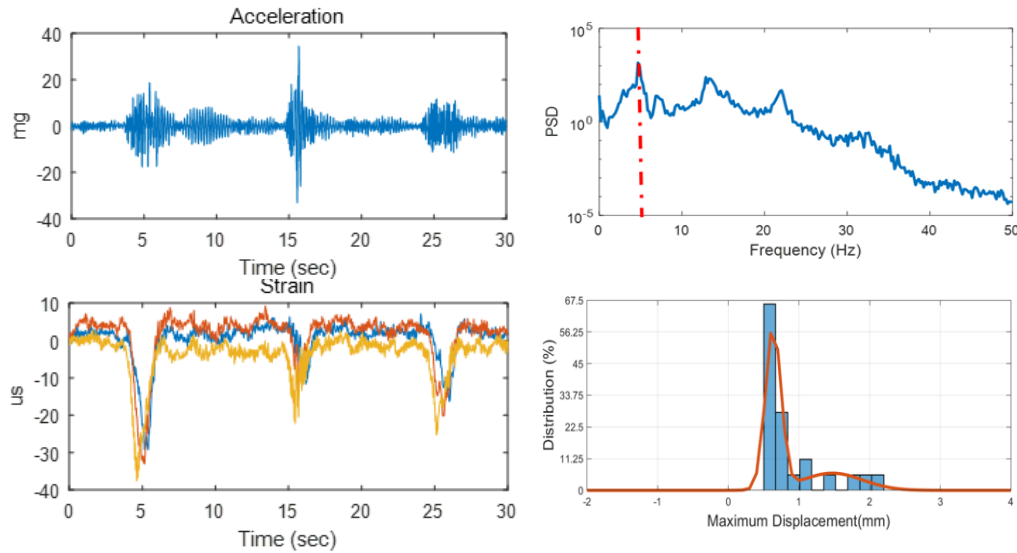
Project ①: IoT Sensing and Cloud System for Smart Bridge Level 3 (MOLIT, 2021.04 ~ 2022.12)

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[Cheongdam Br. No. 1
Installation]



[Measured Data]



LVDT	JANET	Difference
2.08mm	2.18mm	0.1mm
1.01mm	1.09mm	0.08mm
1.26mm	1.31mm	0.05mm

Research Projects (1)

Project ①: IoT Sensing and Cloud System for Smart Bridge Level 3 (MOLIT, 2021.04 ~ 2022.12)

- In addition, to identify and resolve problems arising during field deployment, the system was applied to in-depth safety inspections of 10+ real bridges as of 2022 (20+ bridges as of 2024)
- This revealed many improvements not visible at the lab level, and the sensor's circuit design (hardware) and software were updated 5+ times to enhance performance.

1. Installation Efficiency

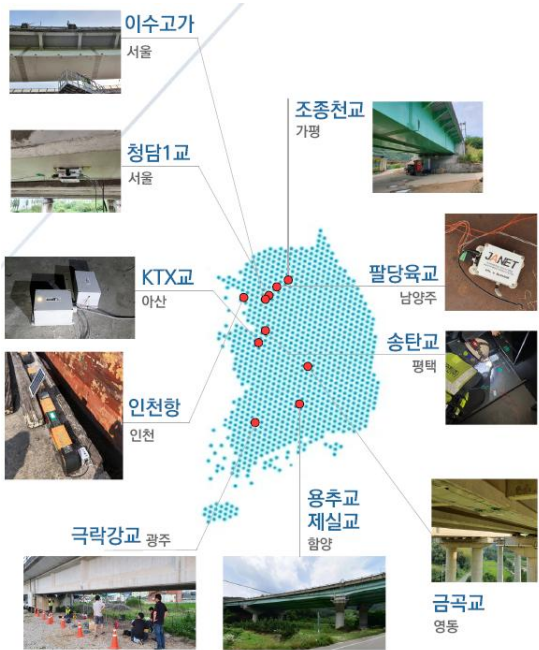
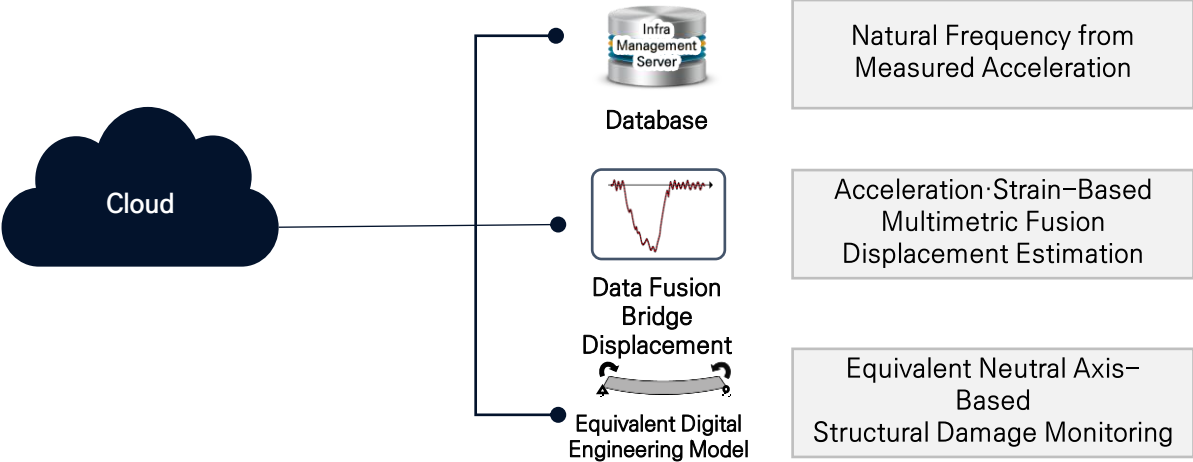
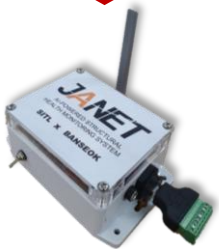
One sensor per physical quantity; cost grows with the number of measured quantities

2. Power Issues

Power management needed for long-term monitoring without battery replacement

3. Communication Issues

Data loss, network connection failures, etc.
Communication error handling



Project ①: IoT Sensing and Cloud System for Smart Bridge Level 3 (MOLIT, 2021.04 ~ 2022.12)

- **Research outcomes** were prepared for new-technology registration over 1+ year after the project ended so they could be used in real bridge safety inspections; as a result, the technology was **designated New Construction Technology No. 959**
- Experiencing firsthand the journey from my first research proposal being selected to its results being registered as a New Construction Technology, I learned **what processes and efforts** are needed for academic technology to be applied in industry



01 현장실사

현장실사 대상 시설물

노선	교량명	종별	준공년도	구조형식	연장	교폭	설계하중	관리주체
6호선	팔당육교	1종	1998년	상부구조(STB거더)	450m	19.5m	DB-24	외정부국토관리공사무

팔당육교 (양수방향) S1

1 신장기술 2 LVDT 3 OU-TYPE

현장 실사

- 01 JANET 센서와 변형률 설치 및 결합상태
- 02 계측결과 무선통신 상태(블루투스)
- 03 실시간 변위 계측 시스템 작동 확인(계측변위)
- 04 처짐 결과값 확인(계측변위와 Reference 측정값 비교 정확도 확인)

반석안전 주식회사

가속도와 변형을 기반 교량의 연직변위 무선측정 시스템 | 16

01 현장실사

현장실사 결과

신장기술	기존기술(LVDT)	기존기술(OU-TYPE)
Case-1(정재하)	Case-1(정재하)	Case-1(정재하)
Case-5(동재하, 60km)	Case-5(동재하, 60km)	Case-5(동재하, 60km)

신장기술과 기존기술 측정계측 값 비교결과 95% 이상의 정확도 확인

반석안전 주식회사

가속도와 변형을 기반 교량의 연직변위 무선측정 시스템 | 17

건설신기술

실시간 100Hz로 데이터 계측 데이터 전송 '특징'

■건설신기술 959호-교량 연직변위 무선 측정시스템

천세윤 기자 | 기사입력 2023/05/11 [07:43]

가속도 변형률 0.3mg 1με 수준 계측분해 성능 '확보'
산정된 연직변위는 오차율 5% 미만 측정 정확도 자랑

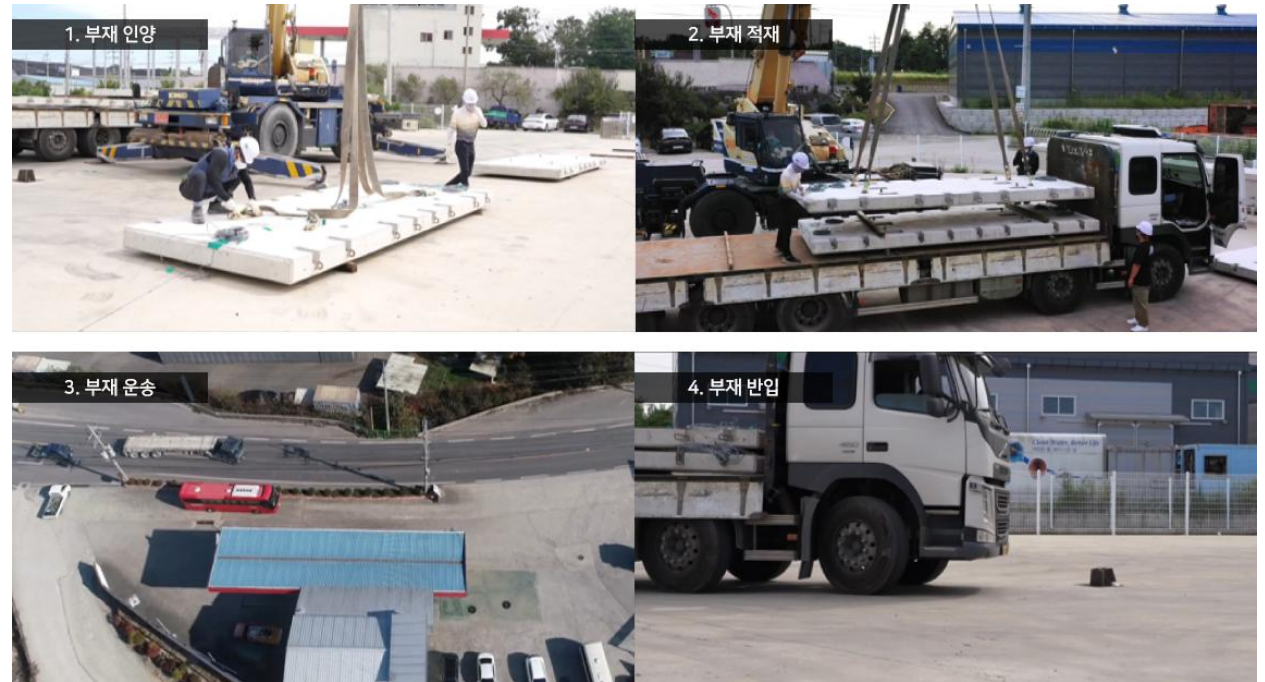
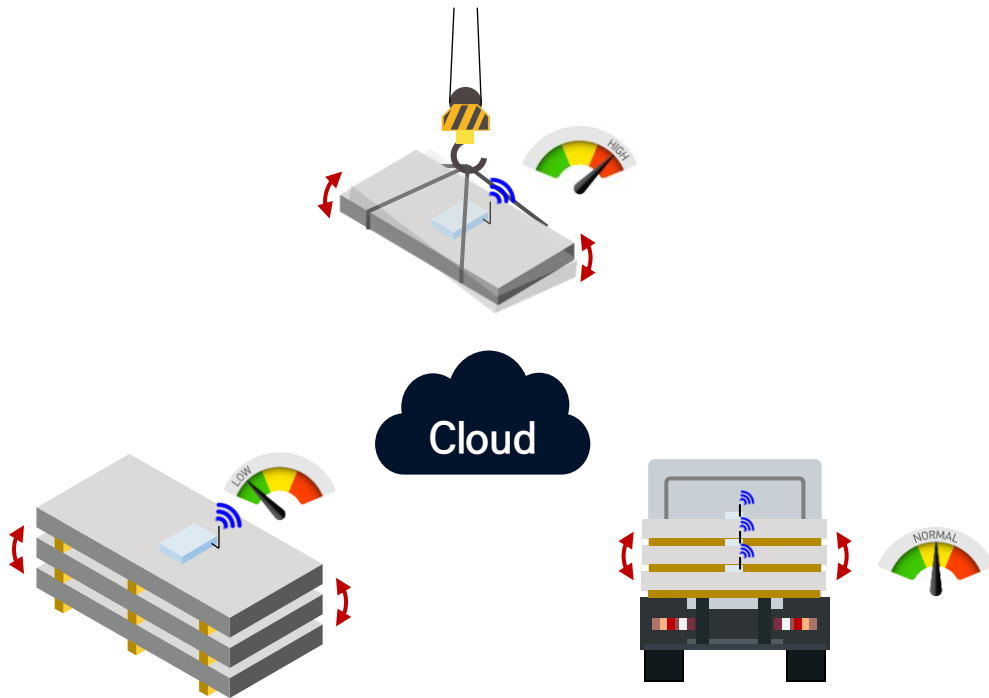
JANET Sensor

□ 3축 가속도 및 3채널 변형률 측정, 블루투스 지원(무선통신), IP67 방수방진

Research Projects (2)

Project ②: Intelligent Road Structure Construction Quality Management (MOLIT, 2020.04 ~ 2023.12)

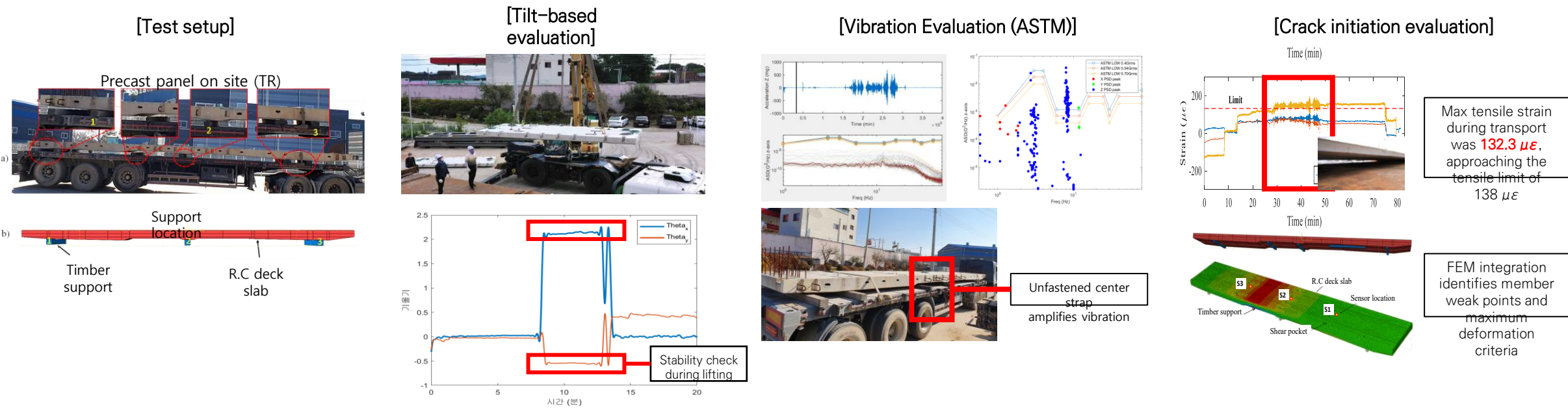
- Contrary to the common belief that factory-made members are safe, **prefab members can also suffer excessive cracking or deformation during transport**, degrading their quality, yet **no technology existed to monitor members in transit**
- JANET was installed on each member** to **collect acceleration and strain data** from prefab members throughout all transport stages (lifting-loading-transport-delivery), and the data were analyzed to **assess safety**



Research Projects (2)

Project ②: Intelligent Road Structure Construction Quality Management (MOLIT, 2020.04 ~ 2023.12)

- To keep prefab members free from **excessive deformation or vibration** during transport, **dunnage must be well supported at the planned points and straps must be properly tightened**
- For each transport stage of prefab members, **tilt, vibration, and strain data were analyzed** to assess safety, an **early-warning system for excessive anomalies was built**, and its **threshold criteria were designed**.
- In collaboration with Bricon Co., Ltd., which operates a prefab R&D plant, **sensors were attached to actual prefab members and trial runs were conducted** to collect and analyze data



Project ②: Intelligent Road Structure Construction Quality Management (MOLIT, 2020.04 ~ 2023.12)

- The research scope expanded from deck slabs to **concrete girders**, performing **monitoring during transport** by attaching JANET sensors and collecting/analyzing data from actual transport

[Pre-installation of JANET]



Flattening the concrete surface with a grinder



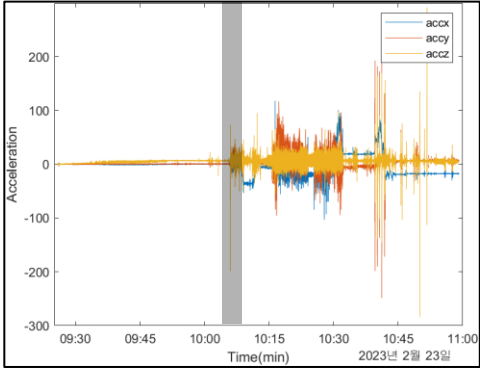
Attaching strain gauges to the prepared concrete surface



Applying adhesive at gauge locations
Attaching steel plate to mount JANET



Connecting JANET extension terminal to 3-channel strain gauges



[Girder monitoring in transit]



① Lifting



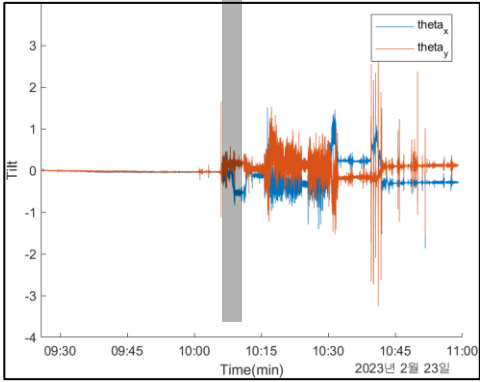
② Loading



③ Transport



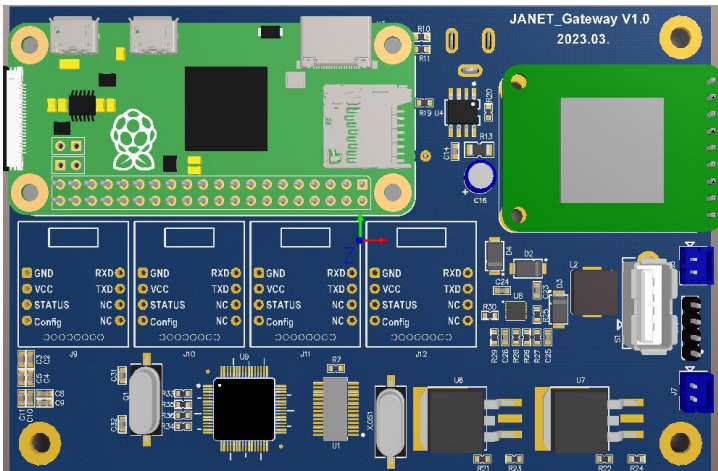
④ Erection



Research Projects (2)

Project ②: Intelligent Road Structure Construction Quality Management (MOLIT, 2020.04 ~ 2023.12)

- To remotely visualize the transport status of prefab members, an LTE-based JANET Gateway circuit was designed
- Data from multiple JANETs attached to each member are collected in real time (via Bluetooth), and with an onboard GPS module, each node's measurements and transport status are sent to the cloud server in real time for management
- Through 4+ field deployments in 2023, operational issues were resolved, sensor installation time was optimized, and the process was improved to make the technology highly field-applicable



JANET Gateway PCB Layout



Packaged JANET Gateway

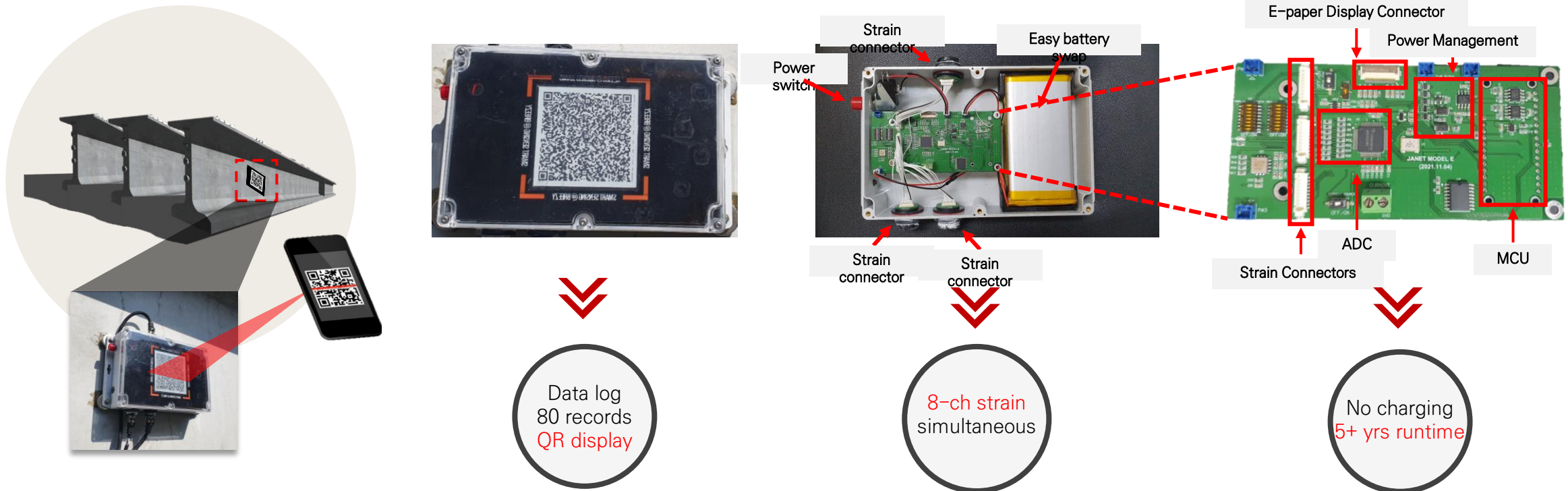


Gateway mounted on trailer

Research Projects (3)

Project ③: Digital Design–Fabrication–Construction Support for Road Structures (MOLIT, 2020.04 ~ 2023.12)

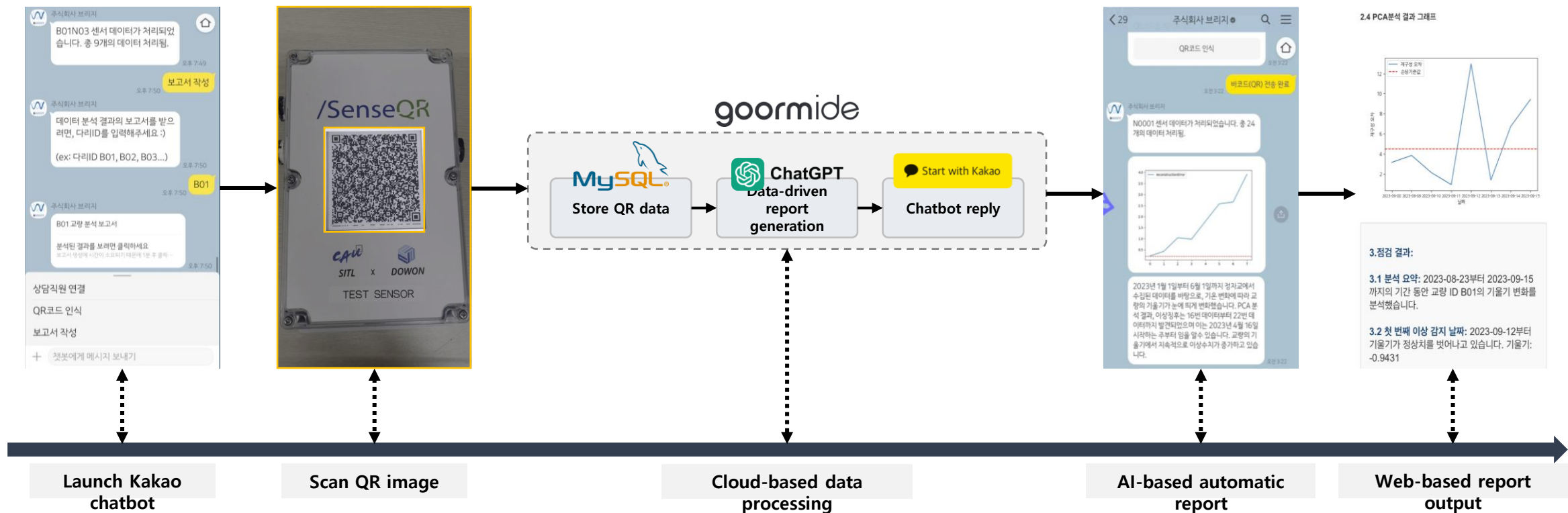
- A sensor was required for **long-term measurement of absolute (static) strain** over the **entire life cycle** of prefab members.
- To run for **10+ years** without battery replacement or charging, an ultra-low-power circuit with **0.34 mA** current draw in Idle (power-saving mode) was designed; based on a Real Time Clock (RTC), the sensor is **automatically triggered once a week** to perform **multi-channel strain measurement**, and **data were acquired from actual prefab members**.
- **Accumulated data** are displayed as a **QR code on E-paper**, designed so that **drones/robot dogs** can in the future **automate the data collection process**



Research Projects (3)

Project ③: Digital Design-Fabrication-Construction Support for Road Structures (MOLIT, 2020.04 ~ 2023.12)

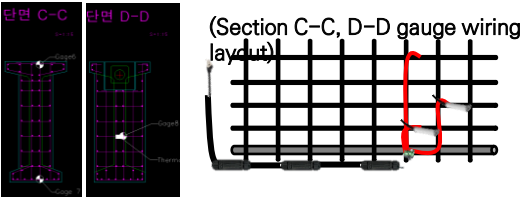
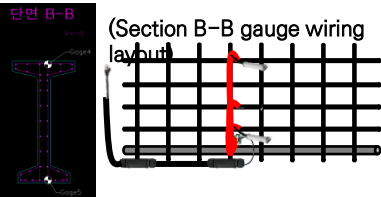
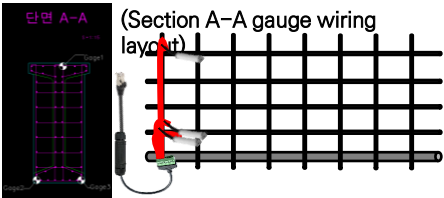
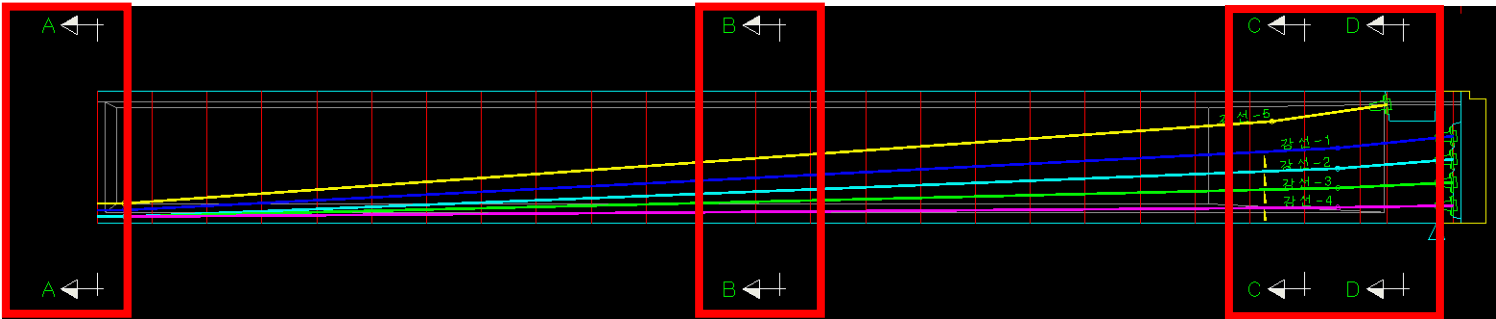
- Using a KakaoTalk chatbot, requiring no separate app development or installation, a data collection module was designed; through the smartphone's LTE connection, data reach the server without a dedicated communication module on the sensor.
- Scanning the QR code displayed on the sensor automatically decodes the data and stores it in the DB, and a built-in GPT-based prompt instantly generates a Data-Driven report



Research Projects (3)

Project ③: Digital Design-Fabrication-Construction Support for Road Structures (MOLIT, 2020.04 ~ 2023.12)

- From Nov 2022 to Jul 2023, inside the girders of **Chohacheon Bridge** (then under construction in Gyeonggi-do; sensors retrieved after completion), **8 embedded strain gauges** and a **1-channel thermometer** were installed, and data were collected at regular intervals using the QR sensor



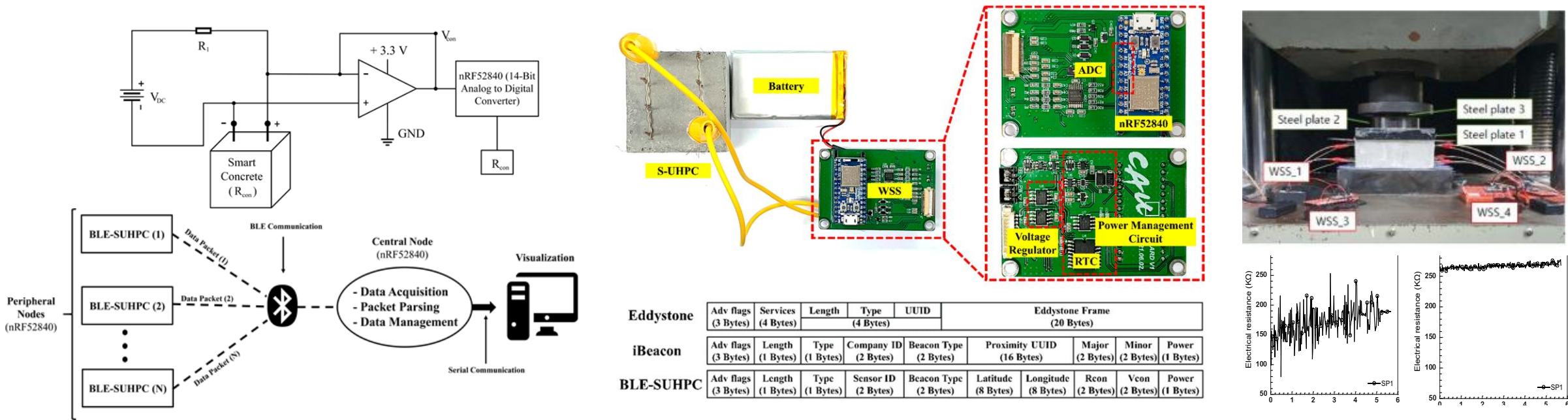
Chohacheon Br.
Site photos



Research Projects (4)

Project ④: Basic Research Lab for Bridge Condition Assessment Using Digital Twins (MSIT, 2019.09 ~ 2022.02)

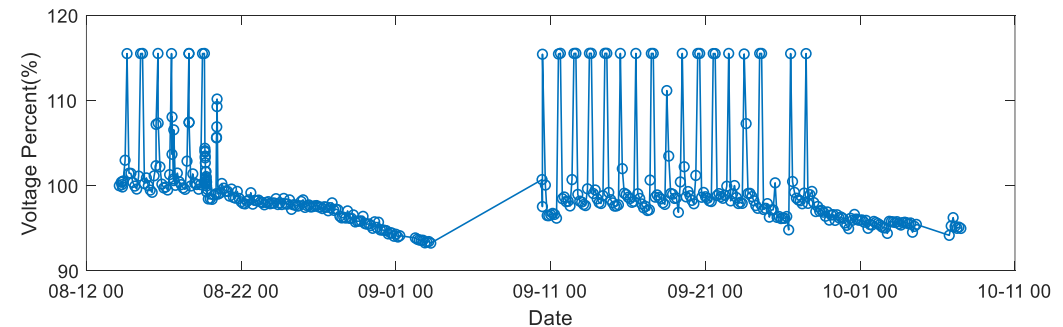
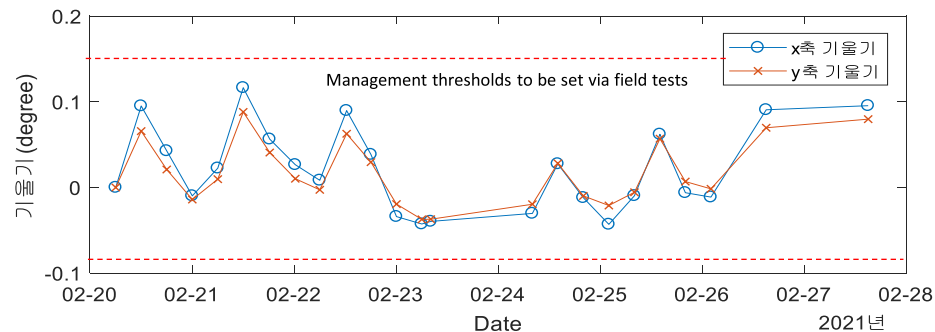
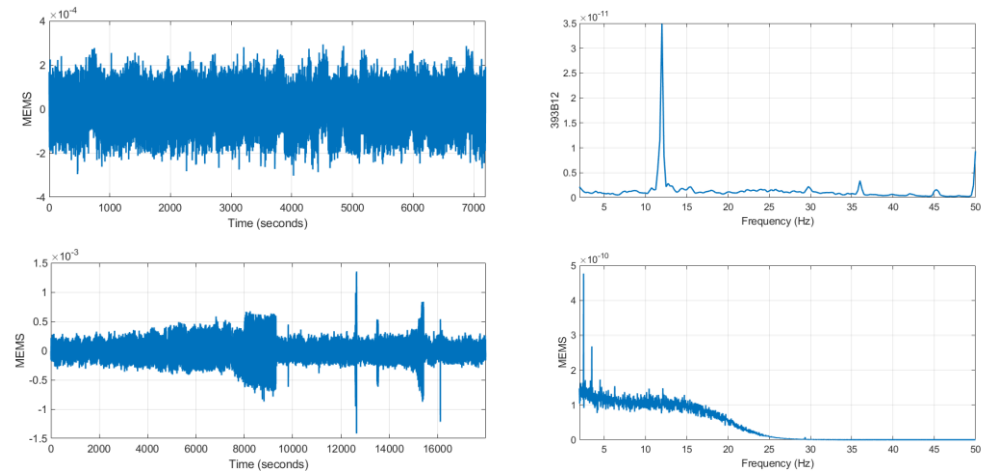
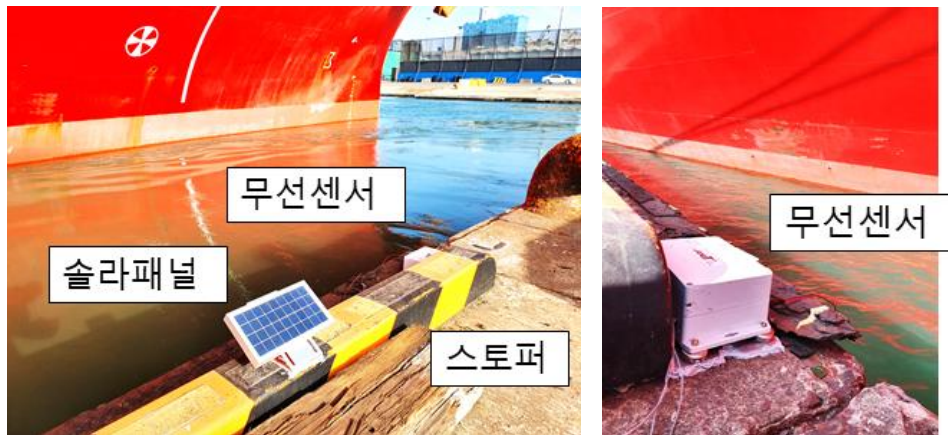
- Applying **smart concrete** between the structure and the strand anchorage and acquiring its data enables **proper detection of structural stress**
- A measurement system was developed to **measure resistance change from voltage change** in smart concrete blocks and to identify the relationship between tension/compression and the resulting voltage change
- The developed smart concrete sensor uses **Beacon**, a low-power Bluetooth protocol, so that a single Master device can receive measurements from multiple concrete sensors, and collected data are logged/visualized via a Python server
- Experimental conditions such as power supply and electrode connection were modified, and a noise filter (Median Filter) was applied to minimize the effects of noise and impulses



Research Projects (5)

Project ⑤: Port Monitoring System Development (KICT contract, 2021.03 ~ 2023.11)

- Monitoring and analyzing the behavior and condition of port facilities requires measurement technology that can operate in extreme salt-damage and operating environments
- An IP67-rated waterproof case was used to package JANET, and solar panels enable self-charging, allowing monitoring of port tilt, vibration, and natural frequency changes.
- Sensor data are transmitted wirelessly to the cloud server via a built-in LTE-M modem, and a Watch Dog Timer (WDT) is embedded for Fault-Tolerant operation

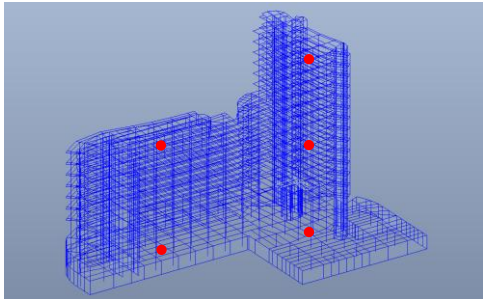


Research Projects (6)

Project ⑥: TCP/IP-Based Sensor Framework for Continuous Building Vibration Measurement (Soongsil Univ. contract, 2022.03 ~ 2024.02)



[Hyunnam Eng. Building]



[MIDAS Model]



- Soongsil University's Hyunnam Engineering Building consists of **two connected reinforced-concrete structures**, one with 9 floors above ground and the other with 15, forming an **asymmetric structure**
- To **analyze the dynamic characteristics of each building** in the asymmetric structure, JANET was installed at 5 locations, and to obtain time-synchronized data, data were collected via TCP/IP over the campus network



[Node 1-1]



[Node 1-2]



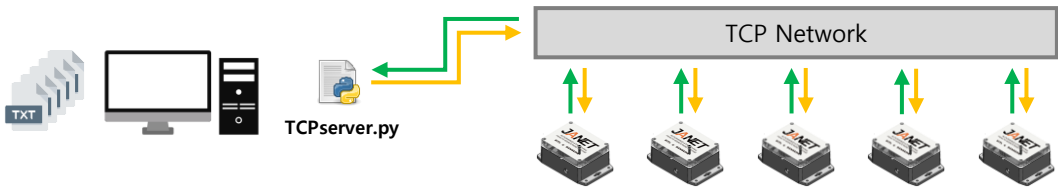
[Node 2-1]



[Node 2-2]

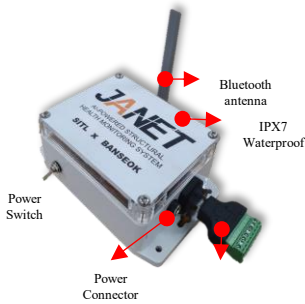


[Node 2-3]

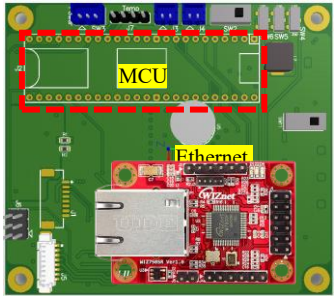


[TCP/IP network diagram]

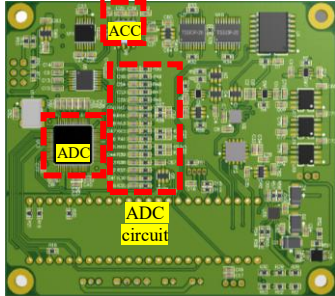
- Built a **Python-based multi-port data acquisition server** over TCP/IP communication
- Flask-based trigger automatically extracts **peak acceleration** and **FFT-based natural frequency** for each 10-minute window



[IoT Sensor JANET]



[Sensor front]



[Sensor rear]

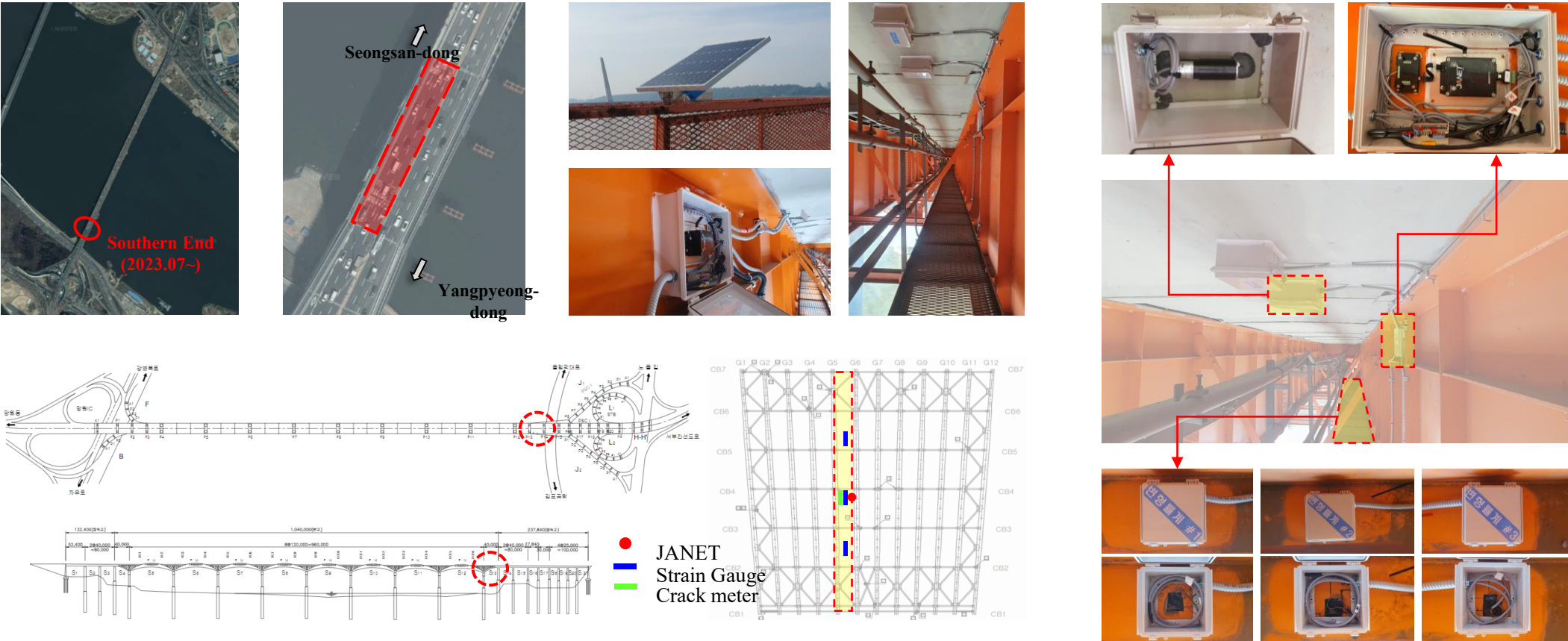


- Precision within **0.1 mg**; triaxial acceleration sampled at **100 Hz** for precise analysis of structural dynamics
- An Ethernet modem (WIZ750sr) with **10/100 Mbps Ethernet / max 230 kbps serial** is built in, transmitting **100 Hz triaxial acceleration data in real time** for building monitoring

Research Projects (7)

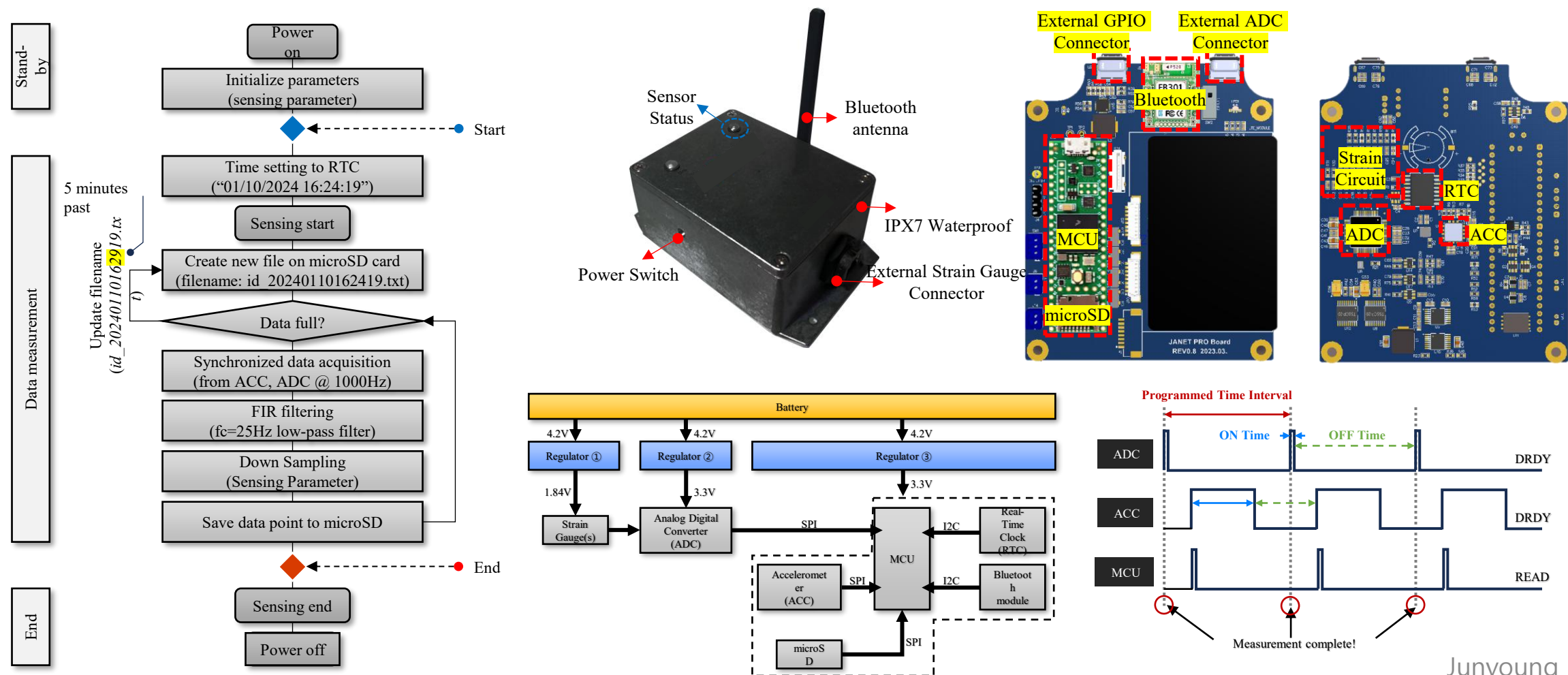
Project ⑦: Seongsan Bridge Safety Inspection – Online Mid-Term Monitoring (Seoul Metropolitan Gov., 2023.01 ~ 2024.02)

- JANET was installed on the south/north ends of Seongsan Bridge to monitor the bridge's ambient vibration/strain/displacement and deck vibration/cracks.
- Sensor data are stored in a cloud DB via an LTE modem and managed through a web-based dashboard



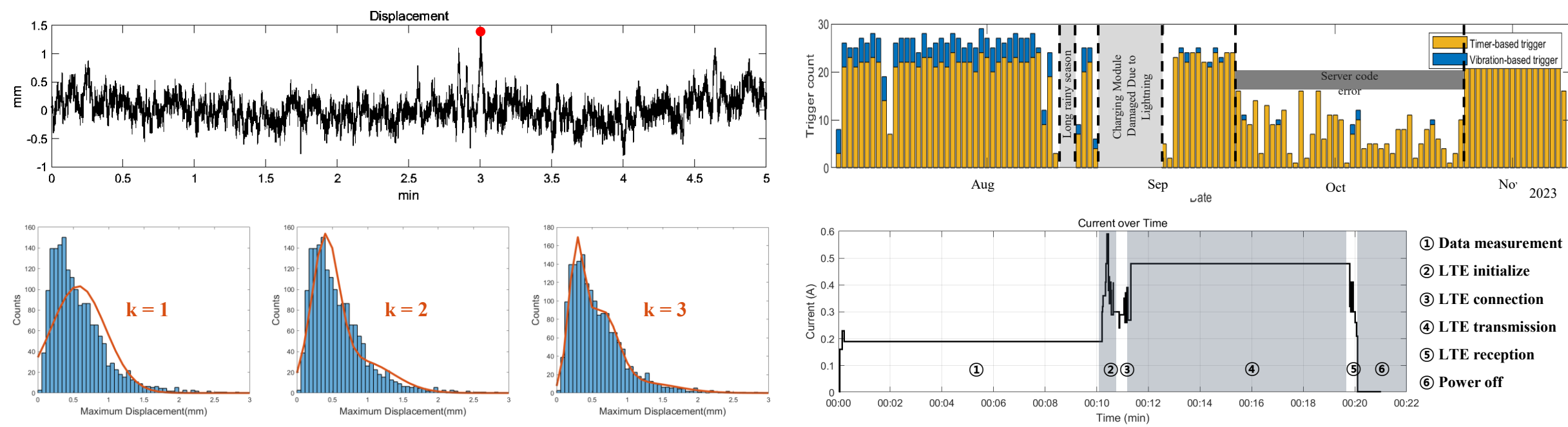
Project ⑦: Seongsan Bridge Safety Inspection – Online Mid-Term Monitoring (Seoul Metropolitan Gov., 2023.01 ~ 2024.02)

- Improved the circuit stability of the existing sensor, strengthened its low-power design, and redesigned the PCB and enclosure for mass production
- Replaced the analog accelerometer (ADXL345) with a digital accelerometer (ADXL355), expanding analog measurement channels from 5 ch to 8 ch



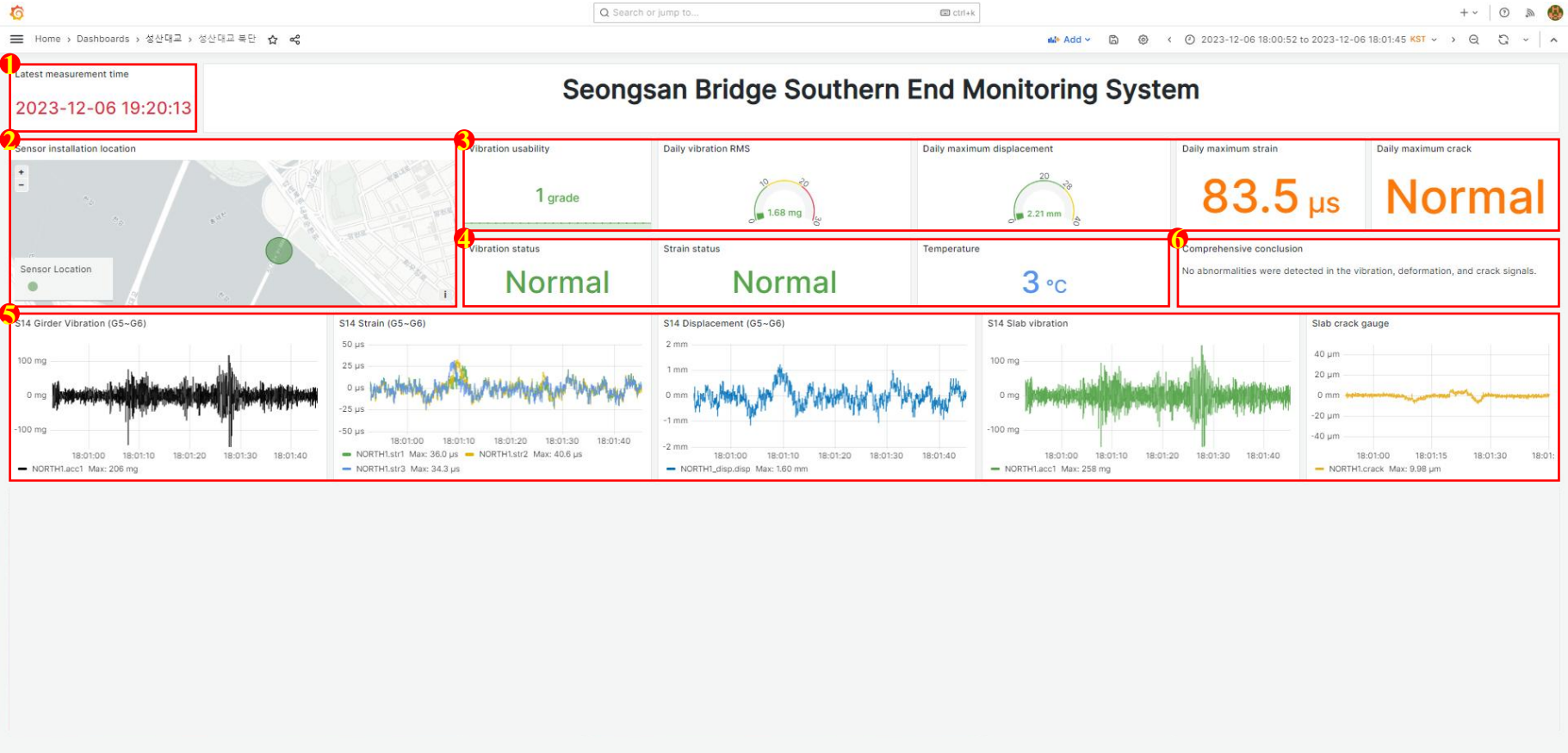
Project ⑦: Seongsan Bridge Safety Inspection – Online Mid-Term Monitoring (Seoul Metropolitan Gov., 2023.01 ~ 2024.02)

- Using JANET on Seongsan Bridge, displacement history from long-term monitoring was analyzed, confirming that the maximum displacement distribution stayed within the bridge's allowable deflection ($L/1000$), on the safe side
- Following data loss during operation, the server code was improved and the solar panels were inspected to enhance stability
- In addition, natural frequency and crack-gauge change histories were continuously monitored from long-term data



Project ⑦: Seongsan Bridge Safety Inspection – Online Mid-Term Monitoring (Seoul Metropolitan Gov., 2023.01 ~ 2024.02)

- Entering the server address:port gives access to the live server; the visualized **web dashboard** looks as follows

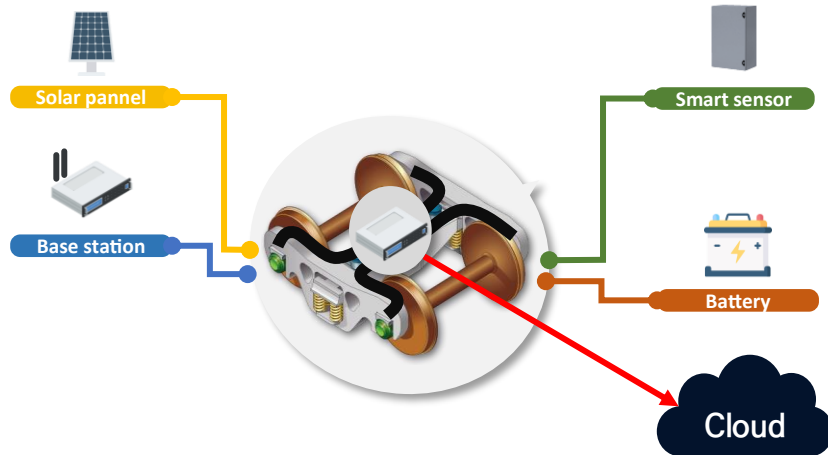


- ① Last sensor measurement time
- ② Sensor installation location
- ③ Vibration grade, RMS, displacement grade, strain status
- ④ Vibration/deformation/temperature status
- ⑤ Time-series data visualization
- ⑥ ChatGPT-based overall assessment

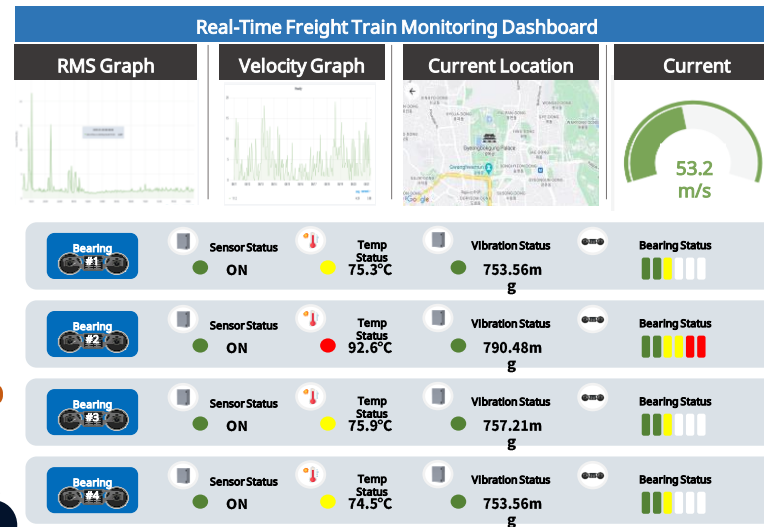
Research Projects (8)

Project ⑧: Freight Train Monitoring System (freight car maker contract, 2023.11 ~ 2024.02)

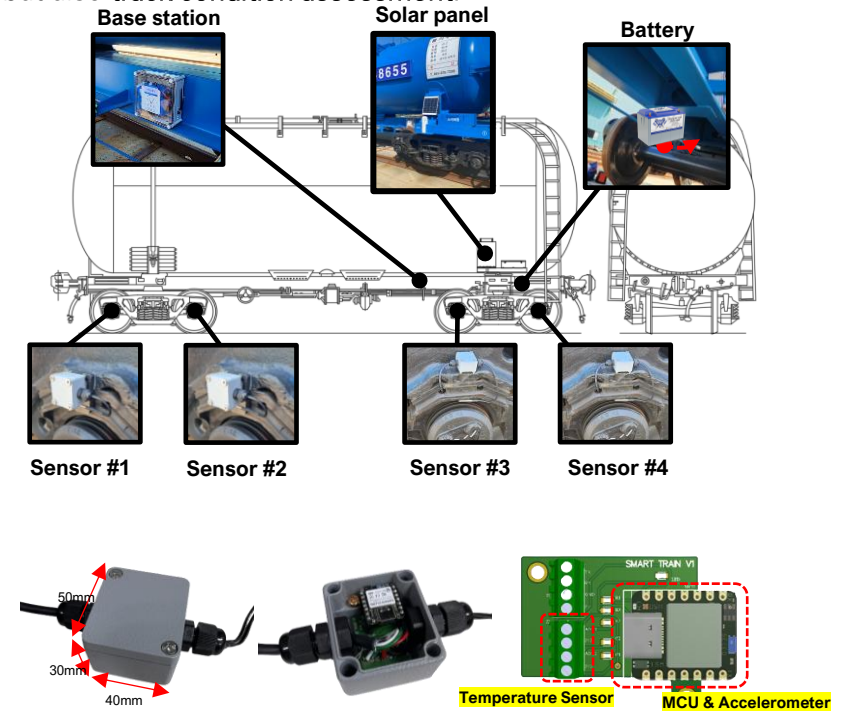
- Freight train (freight car) bearings are assessed only by visual inspection, so latent **fatigue damage** or **deformation damage** caused by high speeds and heavy loads goes undetected
- Accordingly, an ultra-compact sensor attachable to each bearing was developed to **monitor bearing vibration and temperature**, forming a freight car monitoring system that sends data to the server via LTE
- The system is **collecting data on actual in-service freight cars**, and by improving stability and expanding deployment, aims to **develop algorithms for diagnosing freight car bearings**
- The scope also includes learning **GPS-based, location-dependent anomaly patterns** to perform not only bearing diagnosis but also **track condition assessment**.



1. Smart Sensor

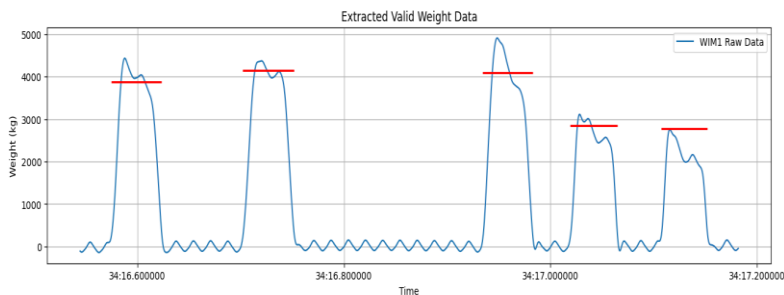
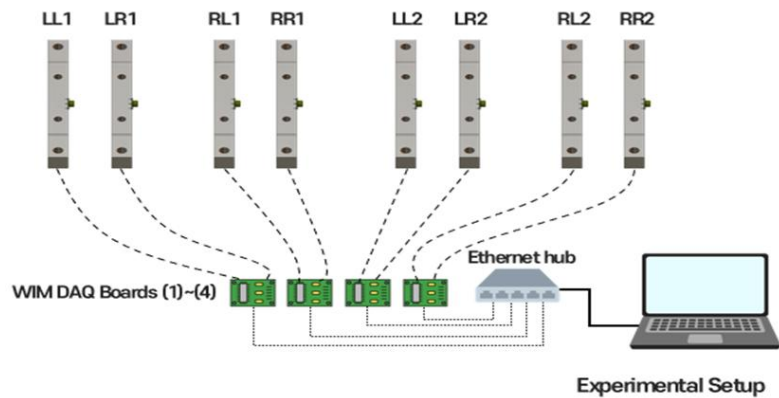


2. Cloud System



Project ⑨: AI Data-Driven Technology for Improving Freight Truck Transport Safety (MOLIT, 2025.01 ~ present)

- Designed and developed a DAQ (Data Acquisition) board for WIM to replace NI equipment in the AI-based freight truck transport safety project
- Designed a multi-channel analog acquisition circuit with high-speed sampling (50 kHz), implemented filtering and downsampling (5 kHz), and optimized the measurement system with an 18-bit ADC
- PoE-based power design and load cell integration reduced enclosure size and improved installation/operation efficiency; designed a modular strip-type WIM system architecture
- Redesigned scale factors and weight calculation algorithms for measurement differences versus the legacy NI DAQ, and validated performance with field driving data



WIM DAQ 보드 단위 변환

1. 비트 → 전압 변환

- AD7989 입력 전압 [V] = 비트 값 [bit] × (인가 전압 / 비트 분해능) [$\mu\text{V/bit}$]

$$V_{DAQ} = N_{bit} \times (V_{EXT} / 2^N)$$

152.58789 $\mu\text{V/bit}$

LSB means least significant bit. With the $\pm 5\text{V}$ input range, 1 LSB is 152.58789 μV

- 여기서, 인가 전압은 10V(5V Differential), 비트 분해능은 2^{16}
- (ADC는 18비트이지만 바이너리 전송 효율을 높이기 위해서 2바이트(16bit)만 스케일 다운해서 사용, 2비트 버림)

2. 증폭기 Gain 보정

- 로드셀 출력전압 [V] = AD7989 입력 전압 [V] / Gain

$$V_{loadcell} = V_{DAQ} / \underbrace{G_{amp}}_{99}$$

3. 풀스케일 비율 계산

- 풀스케일 비율 = 로드셀 출력전압 [V(V)] / 로드셀 감도 [V/V]

$$R_{fullscale} = V_{loadcell} / \underbrace{S_{loadcell}}_{0.003 \text{ V/V}}$$

4. 최종 중량 변환

- 최종 중량 [kg] = 풀스케일 비율 / 로드셀 정격용량[kg]

$$W_{final} = R_{fullscale} \times \underbrace{W_{fullscale}}_{22,200 \text{ kg}}$$

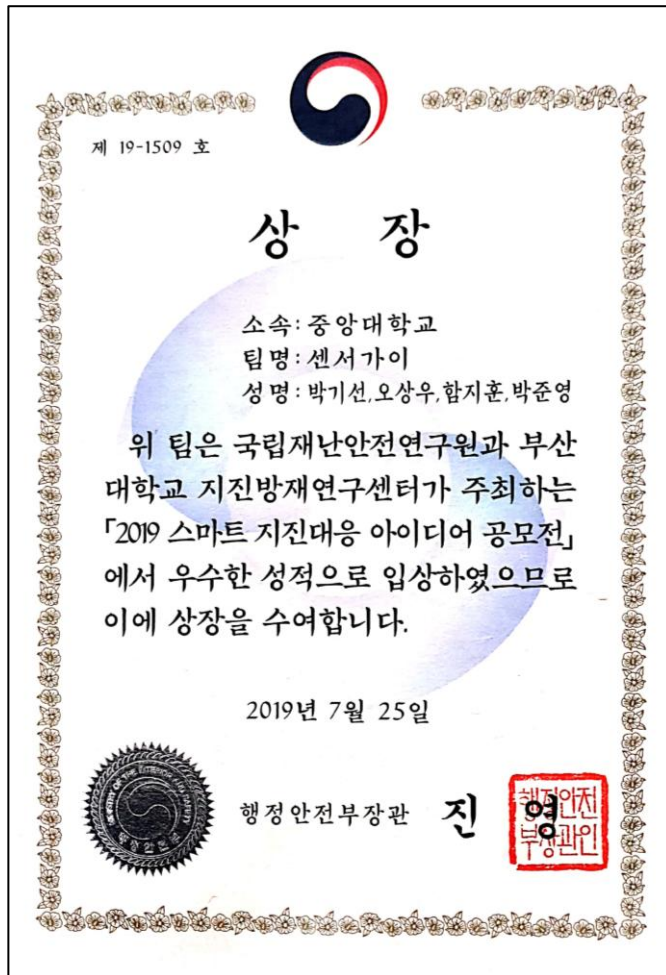


$$\begin{aligned} > W_{final} &= \left(\frac{N_{bit} \times (V_{EXT} / 2^N)}{5V} / \text{Gain} \right) \times (W_{fullscale} / S_{loadcell}) \\ &= N_{bit} \times (152.58789 \mu\text{V} / 5 / 99) \times (22200 / 0.003) \\ &= N_{bit} \times 2.281112 \end{aligned}$$

$$> \text{스케일 팩터} = 2.281112 \text{ [kg/bit]}$$

Key Awards

Key Award (1)



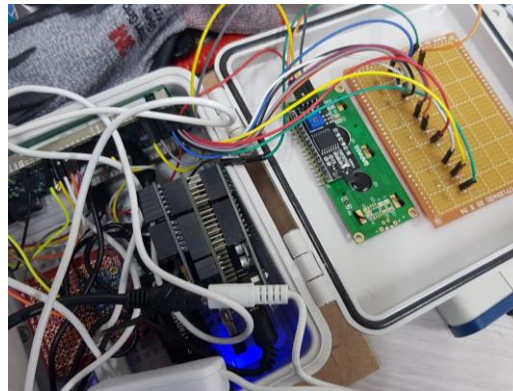
❖ Title: Earthquake Alert System Using Multisensory Media and Smart Sensors (2019)

❖ Overview

- Developed a **real-time earthquake detection algorithm** that recognizes excessive vibration upon an **STA/LTA trigger** and distinguishes seismic, random, and sine waves within 10 s
- Developed an **ultra-low-power BLE smart seismic sensor** with the detection algorithm embedded
- Developed an **alarm propagation system** using nearby Bluetooth devices over BLE wireless communication
- Field validation on the shaking table at Pusan National University's Yangsan Campus tested whether 10 input waveforms were correctly classified and alarmed; ours was the only team to detect all 10 correctly, earning the Minister of the Interior and Safety Award

❖ Related article

- https://www.materic.or.kr/mobile_new/rt/content.asp?idx=162383&page=2&board_idx=1031&

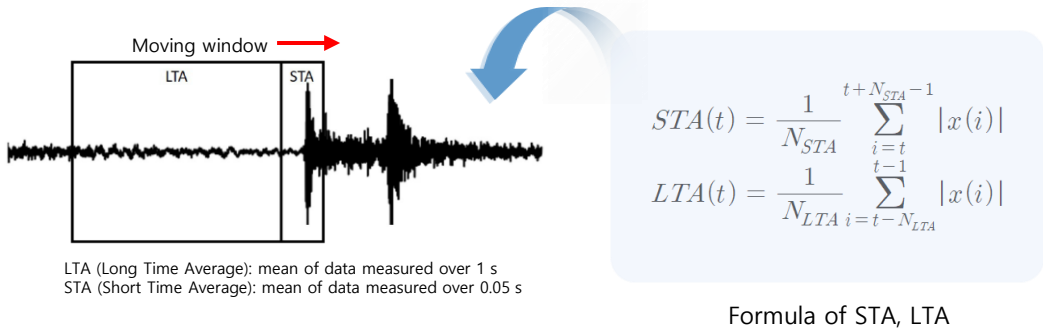


Key Award (1)

❖ Title: Earthquake Alert System Using Multisensory Media and Smart Sensors (2019)

- Proposed real-time earthquake detection (classification) algorithm

1 STA/LTA Algorithm



- Uniaxial (x-axis) acceleration collected at 1100 Hz
- $N_{STA} = 55$ $N_{LTA} = 1100$ set so that LTA is the 1-s data average and STA is the 0.05-s data average

2 Trigger parameter setting

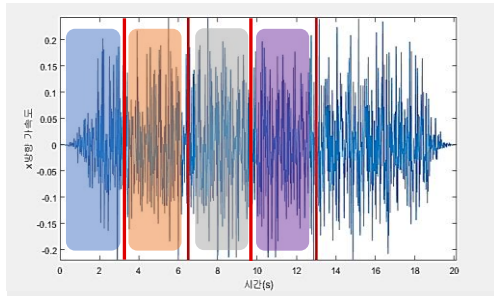
Setting N

$$\alpha < \frac{STA}{LTA}$$

Trigger Formula

- α value determines the moment data are read
- α smaller $\rightarrow$ more sensitive to change; larger $\rightarrow$ less sensitive
- $\alpha = 1.6$ set so that if the 0.05-s acceleration average exceeds 1.6 $\times$ the acceleration average over the past 1 s, a significant change is deemed to have occurred and a Trigger fires

3 Time domain Classification

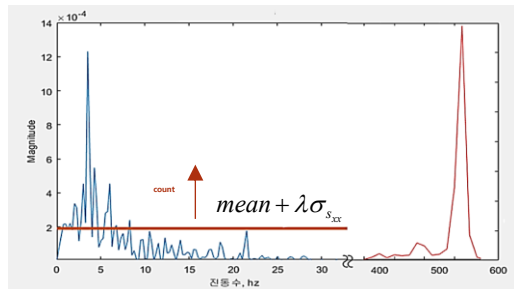


- $$\frac{\min(\sigma_k)}{\max(\sigma_k)} > \tau_{random}$$
- 1100 acceleration time-series samples over 10 s are split into 4 windows for analysis
 - Sine waves are identified by the variance ratio across windows, exploiting their constant vibration trend over time

Parameter settings $\rightarrow$ $\tau_{random} = 0.75$

Sine wave criteria

4 Frequency domain Classification

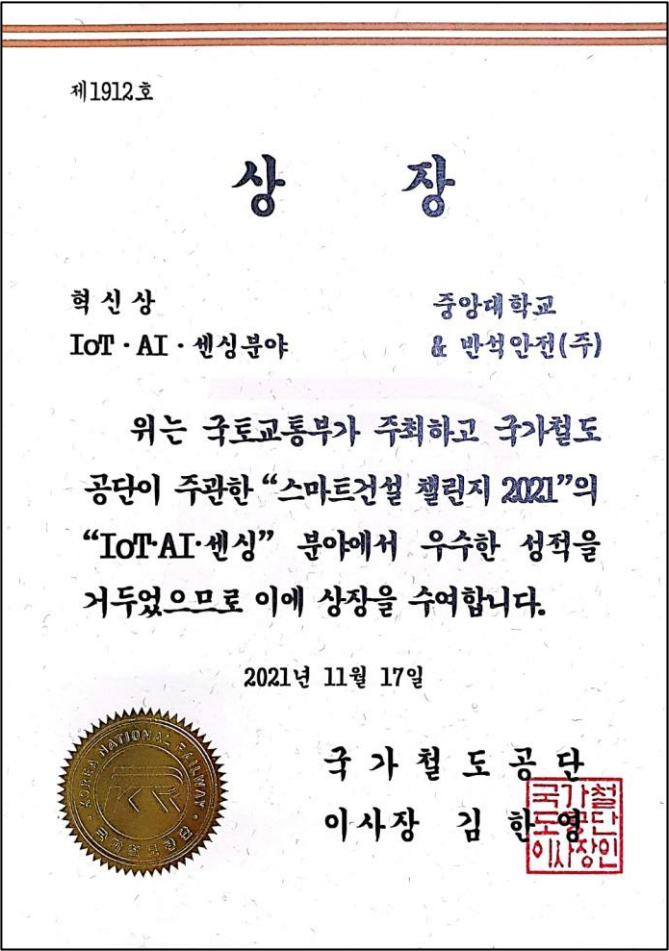

$$S_{TH} = \{v | v > E[S_{xx}] + \lambda \sigma_{S_{xx}}, \forall v \in S_{xx}\}$$

- λ value determines the number of counts
- Building vibration (structure) vs. earthquake distinguished by count

Parameter settings $\rightarrow$ $\lambda = 0.5$
 $n(S_{TH}) > 25$

Quake criteria

Key Award (2)



❖ Title: IoT Sensor and Cloud-Based Continuous Bridge Displacement Monitoring System (2021)

❖ Overview

- Bridge maintenance centered on infrequent in-depth safety inspections struggles to remain sustainable amid a **shrinking construction workforce** and an **increasing number of aging bridges**
- Developed a system that uses **IoT sensors** charged by solar power (no separate power unit) and built-in LTE modems to send data wirelessly to a **cloud server**, enabling **daily detection of the bridge's dynamic behavior**
- From acceleration and multi-channel strain measured by JANET, **displacement is computed using Multimetric Data Fusion**, enabling long-term monitoring of displacement under ambient traffic loads and natural frequencies
- **System reliability was verified on 10+ real bridges**, and results from 3+ months of deployment on Cheongdam Bridge No. 1 (Seoul) and the Paldang Overpass formed the basis of our entry
- Received the award as the **only academic participant in the session**



Key Award (2)

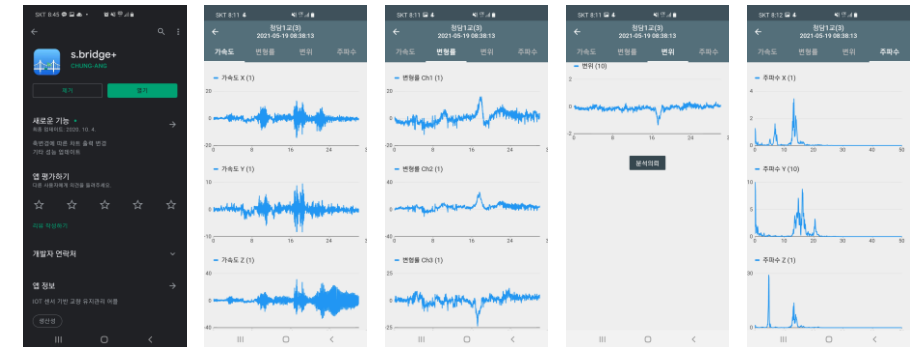
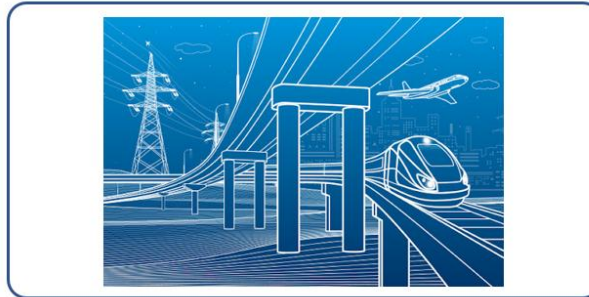
❖ Title: IoT Sensor and Cloud-Based Continuous Bridge Displacement Monitoring System (2021)

- Promoted that JANET can perform both short-term bridge monitoring (for load-carrying capacity evaluation) and long-term monitoring (for digital SOC).
- Since sensor data are collected wirelessly via LTE, we demonstrated that bridges can be managed remotely at all times

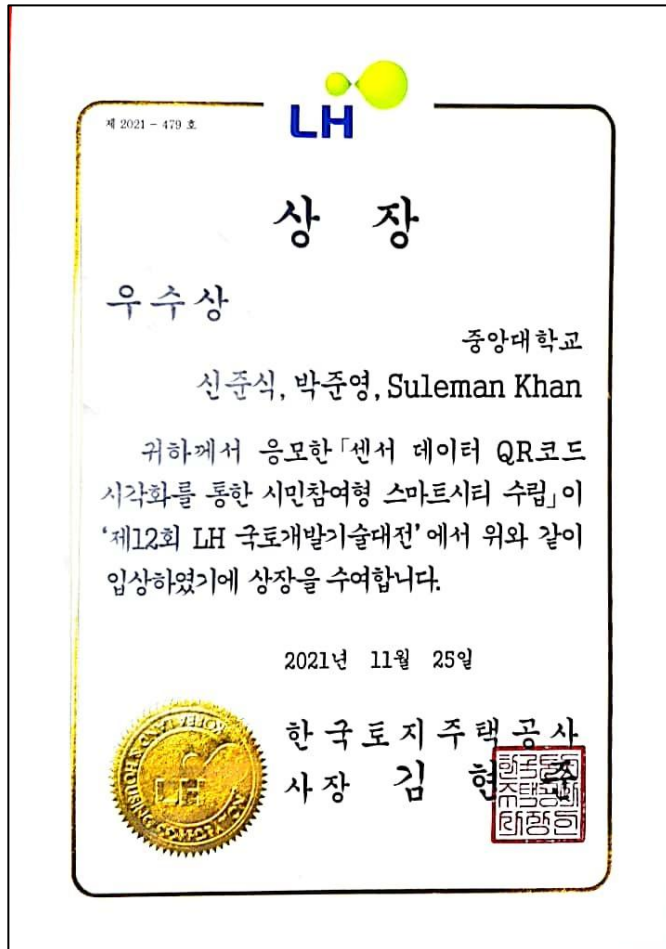
단기 모니터링 : 교량 안전성 관리를 위한 중요지표



장기 모니터링 : 디지털 SOC로 나아가는 시대



Key Award (3)



❖ Title: Citizen-Participatory Smart City via QR-Code Visualization of Sensor Data (2021)

❖ Overview

- Developed an ultra-low-power IoT sensor that collects urban data (tilt, odor, noise, temperature, etc.) and visualizes it as a QR code on an EPD
- Designed a KakaoTalk chatbot-based data collection framework that automatically extracts data from QR codes scanned by smartphones and stores it in a DB
- Proposed a citizen-participatory urban maintenance model in which data are collected through passers-by's 'participation', with no communication module on the sensor

❖ Related article

- https://news.cau.ac.kr/cms/FR_CON/BoardView.do?MENU_ID=10&CONTENTS_NO=&SITE_NO=5&BOARD_SEQ=1&BOARD_CATEGORY_NO=&BBS_SEQ=6508



Key Award (3)

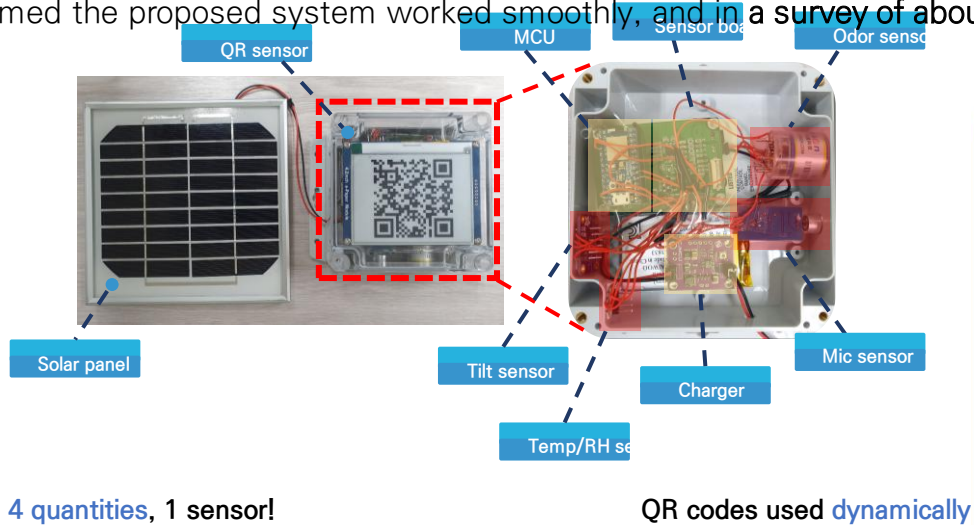
❖ Title: Citizen-Participatory Smart City via QR-Code Visualization of Sensor Data (2021)

- Microphone, IMU, temperature/humidity, and odor sensors with a low-power MCU collect urban environmental data, and E-paper is used to visualize measurement data as QR codes.
- One-day testing confirmed the proposed system worked smoothly, and in a survey of about 20 citizens, over 80% were positive



Aug 28, 2021, Heukseok-dong

[Calibration of developed sensor]

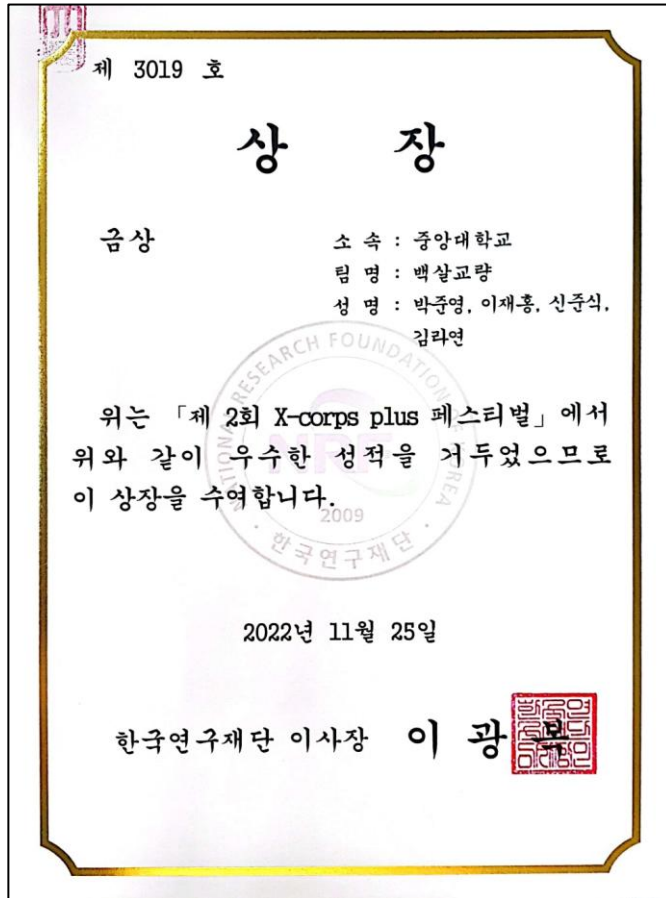


[Ultra-low-power urban sensing sensor]



[KakaoTalk-based QR data collection pipeline]

Key Award (4)



❖ Title: IoT-Based Smart Monitoring System for Stability and Quality Control of Prefab Members in Transit (2022)

❖ Overview

- Built a system to monitor prefab members during transport
- Designed a gateway sensor that uses JANET to measure prefab member vibration (acceleration, tilt) and multi-point strain at 100 Hz, and collects transport data in real time per vehicle carrying multiple prefab members via Bluetooth Classic
- Designed a smartpad that converts dunnage pressure into voltage changes to detect member load distribution and sends a Bluetooth alert when loading problems occur
- Developed a real-time in-transit member safety assessment algorithm from vibration/strain data at each transport stage (lifting-loading-transport) of prefab members
- Developed a cloud-based alarm system that sends real-time KakaoTalk alerts to the safety manager when anomalies occur

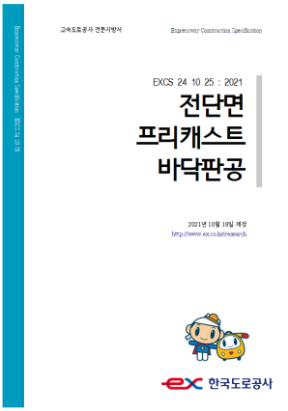
❖ Related article

<http://sit.cau.ac.kr/news.html>



❖ Title: IoT-Based Smart Monitoring System for Stability and Quality Control of Prefab Members in Transit (2022)

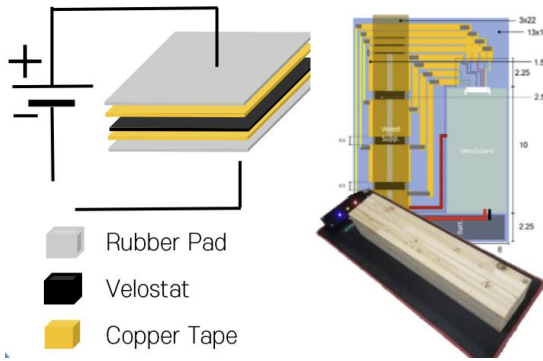
- Per the revised specification guidelines, as a measurement system able to monitor shape changes of precast panels during transport, JANET and a gateway sensor were proposed.
- In addition, to complement detection of proper dunnage support and strap tension, which affect prefab members in transit, a prototype smartpad was developed and proposed



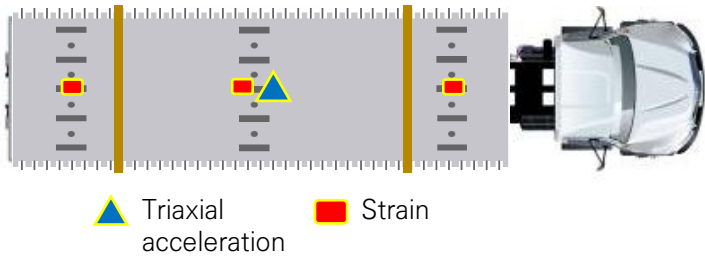
- 3.1.4 보관 및 운반
- (1) 전단면 프리캐스트 패널 하면이 바닥면과 직접 접촉되는 것을 피하고 뒤틀림이 발생하지 않도록 견고한 바닥에 적절한 받침대를 설치하여 보관 및 운반하여야 한다.
 - (2) 패널을 겹쳐 쌓아 보관 및 운반하는 경우에는 지점조건을 고려하여 패널의 동일한 위치에 받침대를 각각 설치해야 한다. 쌓는 패널 수는 최대 10매 이내로 하고 패널 상면은 항상 수평을 유지하여야 한다.
 - (3) 장기간 보관이 필요한 경우에는 노출된 철근의 부식을 방지하기 위한 대책을 마련하여야 한다.
 - (4) 전단면 프리캐스트 패널 설치 이전의 형상을 1회 이상 측정하여 기록·관리하여야 하며, 보관 또는 운반 중 전단면 프리캐스트 패널의 형상변화가 발생한 경우에는 조치 계획을 강구하여 공사감독자에게 보고하고 조치하여야 한다.

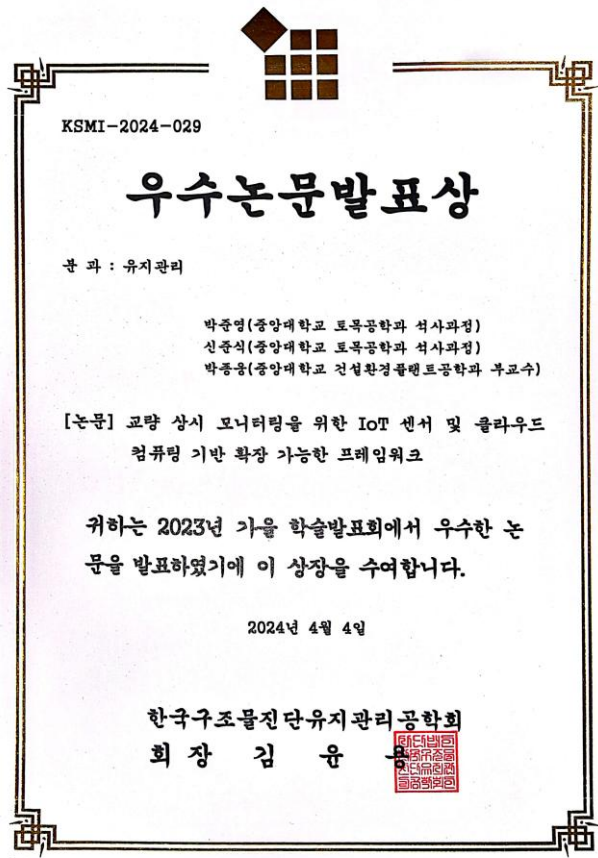


[Monitoring across all transport stages]



[Developed smartpad]





- ❖ Title: Scalable IoT Sensor and Cloud Computing Framework for Continuous Bridge Monitoring (2024)
- ❖ Overview
 - Conventional bridge safety evaluation relies on load testing, but displacement meters require a fixed ground reference, limiting their applicability
 - Acceleration/strain-based indirect displacement estimation has been proposed, but practical use is hindered by the lack of sensors capable of **precise measurement and data synchronization**
 - To address this, an **IoT sensor (JANET PRO)** was developed that **simultaneously measures acceleration (3 ch) and strain (up to 8 ch) at high resolution**, together with an **event-driven low-power system**.
 - Built a monitoring framework that continuously and automatically evaluates bridge safety through cloud-based **data fusion (displacement estimation, vibration characteristic analysis, etc.)**

1. Research Background

□ 한강교량(하천교량)의 변위계측 어려움

- 교량의 안전성 평가는 정적/동적재하시험 이후 **내하력평가**를 통해 그 상태를 평가

변형 및 처짐을 산정

지상 기준점

하천교량의 경우 변위 계측이 어려움

2. Proposed System

□ 데이터처리 알고리즘의 클라우드 자동화

- 변위 연산
- 진동 사용성
- 감각곡선을 통한 진동 사용성 분석

$$U_{total} = \Phi^T \Psi^T \{ \epsilon \} + \frac{1}{\gamma} \Phi \Phi^T \{ \epsilon \}$$

$$Y_s = \sqrt{\frac{S_{yy}^{meas}(w)}{S_{yy}^{meas}(w_s)}}$$

- 고유진동수
- 연속 가속도 CPSD 후 고유진동수 추출
- 온도 API
- 기상정보>성상대고 부근 대기온도

4. Discussion

□ 계측 데이터 분석

- 7월 6일~8월 20일까지의 계측 데이터로부터 최대변위 분포 산출
- 발생 변위와 연직변위 기준치 비교

계측 변위 데이터

최대 변위 분포

경간길이: L=40m
연직변위 기준치: L/2000=20mm

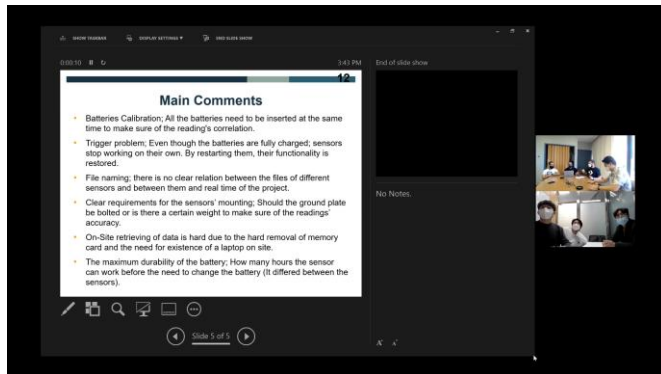
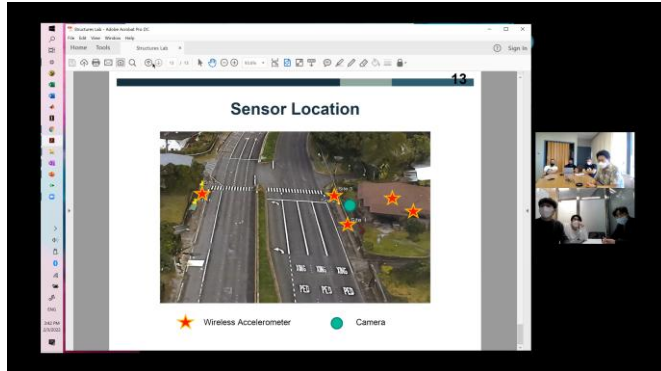
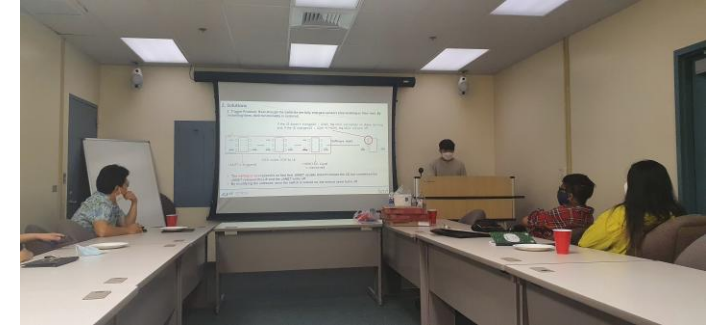
0.1~3.5mm < 20mm

Additional Information

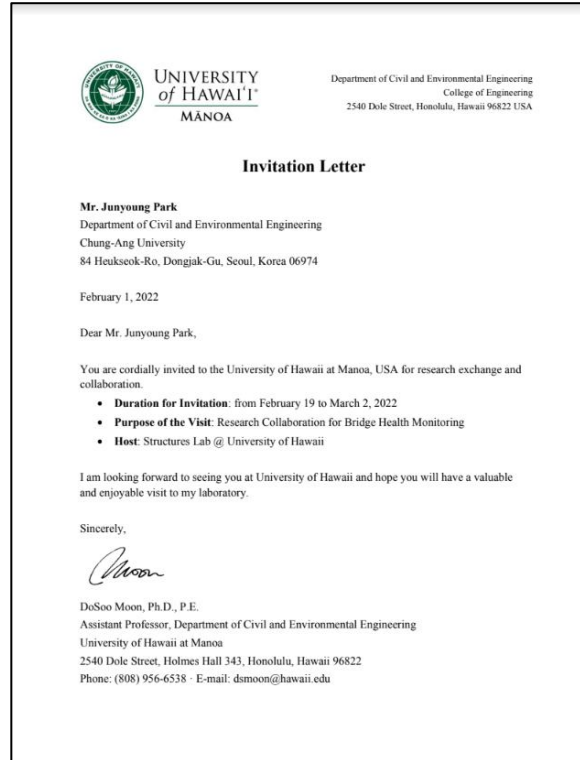
Additional Information (1)

❖ Research Visit to the University of Hawai'i (2022.02.19 ~ 2022.03.03)

- Discussed applications of the JANET sensor developed at Chung-Ang University with the lab of Prof. Dosoo Moon at the University of Hawai'i, shipped sensors to Hawai'i, and shared local deployment results. Subsequently, at Prof. Moon's invitation, I spent 2 weeks in Hawai'i conducting research.
- Gave a seminar on research in Korea and exchanged research topics between the two labs. Also explained sensor fundamentals and debugging methods to local students.



[Zoom Meeting, 2022.02.04]



[Invitation Letter]



[Visit to the University of Hawai'i]

Additional Information (2)

➤ Proposals Written: 16 (R&D 9, Service 7)

Year	Client	Title	Budget	Type	Selected
2026	MOLIT	Freight Truck Weight Information Platform and Field Demonstration for Efficient Overload Enforcement		R&D	✓
2025	MOIS	2025 Automated Road Damage Detection Survey Equipment Purchase Proposal (Draft)		Service	✓
2024	MOLIT	2024–2025 Maintenance of Online Permit System for Restricted Vehicles		Service	✓
2024	MOIS	AI-Based Next-Generation Safety Reporting and Report Value Analysis Service		R&D	✗
2024	NRF	Condition-Based Life Prediction of Freight Cars Using Wheel·Track·Bearing Monitoring and Causal Inference Models		R&D	✗
2024	NRF	Global Basic Research Lab for Meta-Concrete Materials and Structures		R&D	✗
2023	Seoul Metropolitan Gov.	2023 Seongsan Bridge IoT-Based Integrated Monitoring System for Repair Sections		Service	✓
2023	NRF	Next-Generation Cloud Physics Model-Based Vision Displacement Measurement System for Bridges		R&D	✗
2022	KALIS	Study on Automating Safety Diagnosis and Inspection of Infrastructure		Service	✓
2022	KICT	IoT Sensor Measurement and Analysis Using Distance Trigger Source During Ship Berthing		Service	✓
2022	MSS	All-in-One AI Camera System for Automated Bridge Bearing Diagnosis		Service	✗
2022	NRF	Non-Contact Structural Strain Measurement System Based on Super Macro Imaging and Cloud Computing		R&D	✗
2022	NRF	Next-Generation Cloud Physics Model-Based Vision Displacement Measurement System for Bridges		R&D	✗
2021	MOLIT	IoT Sensing and Cloud System for Smart Bridge Level 3		R&D	✓
2021	KICT	Sensor Module for Multivariate High-Resolution Measurement		Service	✓
2021	NRF	Basic Research Lab for Cable Bridge Condition Assessment Using Digital Twins		R&D	✗

Additional Information (3)

➤ Other Awards: 14 (Job-related: 5, Non-job-related: 9)

	Competition	Award	Date	Topic	Awarded by
Job-related	2021 LINC+CAPSTONE DESIGN AWARD	Excellence Award	2021.12.28	Cloud-Based IoT Monitoring System for 3D Displacement Measurement of Bridge Bearings	CAU LINC+ Program
	2021 LINC+CAPSTONE DESIGN AWARD	Popularity Award	2021.12.28	Cloud-Based IoT Monitoring System for 3D Displacement Measurement of Bridge Bearings	CAU LINC+ Program
	8th University Student Water Environment Policy · Technology Contest	Korea Environment Corporation President's Award	2021.10.12	IoT and Deep Learning-Based Smart Water Grid: Focusing on Sewer and Water Supply Pipes	Ministry of Environment
	2020 CAU Engineering Symposium	Dean of Engineering Award	2020.12.10	Bridge Maintenance System Using IoT Sensors	CAU College of Engineering
	3rd X-corps Festival	Encouragement Award	2019.11.29	Integrated Safety Control System for Temporary Structures Using Smart Sensors and IoT	CAU Engineering Education Innovation Center
Non-job-related	2019 12th Jungdok (中讀) Reading Community	–	2019.07.01	2019 Reading Community Activities	Director, CAU Library
	2018 11th Jungdok (中讀) Reading Community	–	2019.03.13	2018 Reading Community Activities	Director, CAU Library
	2018 Fall Course-Linked Capstone Design	–	2018.12.14	Business Model Using an AI Speaker for Engineering	Korea Association of Robot Industry
	2018 'Add MSG to Your Message'	Silver Prize	2018.09.20	New Text Message Service Idea: Ad Lottery	SK Telecom Co., Ltd.
	1st Da Vinci Software Startup Camp	Silver Prize	2018.08.31	franQa: SNS-Style Research App	CAU Da Vinci SW Education Institute
	2018 Engineering Festival	Grand Prize	2018.11.09	AI Object Recognition Hackathon – Guardian: Unmanned Farm Management Drone	Korea Institute for Advancement of Technology (KIAT)
	9th Smart Economy & Management Forum	2nd Place, Team Project	2018.08.26	Real-Time Food Discount Purchasing Service 'Atteul!'	Taihan Fiberoptics, Maeil Business Newspaper
	2018 AI School	Silver Prize	2018.08.14	Team Project	Korea Univ. Engineering Education Innovation Hub Center, KISTI
	KOSPO Future Innovation Idea Contest	Encouragement Award	2018.07.19	Technology and Administrative Systems for a Sustainable Energy Regime and Job Creation	Korea Southern Power Co., Ltd.